MILITARY SPECIFICATION

CHAIR AND STOOL UNIT, DENTAL OPERATING, PORTABLE, IN CARRYING CASE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a portable field dental module (chair and stool unit in a carrying case) for use in the treatment of dental patients under field conditions.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple - Wall.

Military

MIL-C-4150	Case, Transit and Storage, Waterproof and Water-vaporproof.
MIL-M-7298	Manual, Technical, Commercial Equipment.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-151	Metal, Test Methods
FED-STD-595	Color

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of United States Military Property.

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LIST

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24337 Chair and Stool Unit, Dental Operating, Portable

(Copies of specifications, standards, drawings, lists, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES DEPARTMENT OF COMMERCE-
NATIONAL BUREAU OF STANDARDS

Handbook H 28 Screw-Thread Standards For Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. (20402)

3. REQUIREMENTS

3.1 Materials. Materials shall be as specified herein and as specified on the applicable drawings, lists, or specifications designated for the applicable component. When materials are not specifically designated, those used shall be of the highest commercial quality, and entirely suitable for the purpose intended.

3.1.1 Dissimilar metals. Contact between dissimilar metals likely to cause deterioration of parts by galvanic corrosion shall be avoided. Where such contact cannot be avoided, joints or mating surfaces between dissimilar metals shall be protected against galvanic corrosion by plating, coating, insulation, gaskets or other suitable means.

3.2 Drawings. No deviations from the prescribed dimensions and tolerance on the drawings forming a part of this specification are permissible without prior approval of the contracting officer. Any data (such as shop drawings, layouts, flow sheets and processing procedures) prepared by the contractor (supplier) or obtained from a sub-contractor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2.1 Figures. Unless otherwise specified, figures are for illustrative purposes only. Dimensions indicated in figures are in inches. Figures are not drawn to scale.

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3.3 Proprietary components. When trade-named items or other proprietary components are specified, products of other manufacturers may be substituted provided that they are equal to (material, functionally, physically, in quality, performance, and reliability) and interchangeable with the item specified, and prior written approval of the contracting officer has been obtained. The incorporation of an alternate component shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other component unless such modification or change is specifically approved by the contracting officer. Prior to the manufacture of the first article, or if none is required prior to commencing production, the contractor shall submit, for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with the requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, shall not relieve the contractor of the responsibility that these components are interchangeable and will perform in accordance with specified requirements when incorporated into the end product.

3.4 Design, construction, and dimensions. The module shall be a portable dental operating field chair and stool unit in a carrying case suitable for use in the support of providing treatment of dental patients under field conditions. The module shall consist of the following major components: a hydraulically powered, foot operated dental chair, a manually controlled dental stool on casters, and a water-vaporproof reinforced carrying case. (See figures 1 and 2). The carrying case shall accommodate the dental chair and stool in a collapsed, bolted and strapped together configuration as shown in figures 3 and 4. The bottom portion of the carrying case shall also serve as a base for securing the chair when operating under field conditions. The dental chair shall consist essentially of a chair and chair base assembly; the stool shall consist essentially of a seat with an adjustment shaft and base assembly, and the case shall consist essentially of an upper and lower shell. The design, construction and dimensions of the module shall be in strict accordance with the documentation listed in DPSC List 24337. (See Appendix I, tables 1, 2 and 3, for total applicable drawings and parts lists).

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3.4.1 Chair base. (See figure 5). The chair base shall consist essentially of a lift having a floor plate upon which is mounted a series of castings together with other components (such as lift arms and pins) which form a parallelogram which pivots and provides the means to raise and lower the chair. The lifting force shall be developed by a hydraulic power unit consisting essentially of a foot operated hydraulic pump and cylinder assembly, mounted on a common casting. The hydraulic unit shall be self contained and easily removed from the lift for required maintenance. Raising of the lift shall be accomplished by pumping the foot lever (pedal) from a seated or standing position, operable from the back of the chair. Lowering the lift shall be accomplished by depressing a foot operated release lever at the rear of the chair. Lowering speed shall be adjustable by a speed control valve on the hydraulic power unit. The foot pedal lever extension shall be removable and shall clamp under the seat when the unit is in the storage position. The lower part of the lift mechanism shall be protected by a removable cover (housing).

3.4.2 Chair. (See figure 1). The chair shall be sturdy and stable, and shall provide proper support for the body weight during repeated use. The chair (back, seat, and toe section) shall be vinyl upholstered. The top portion (patient contact portion) of each section shall be cushioned with canvas backed foam. The seating portion of the seat section shall be cushioned with a foam pad in addition to the canvas backed foam. The chair shall consist of a pivot casting which shall support the chair seat in such a manner that shall allow it to be tilted in two positions with respect to the floor plate. The “DOWN” position, (Hole B, seat adjustment support, figure 6) shall be used primarily for storage in the case. The “UP” position, (Hole A, seat adjustment support, figure 6) shall be used when a patient is in the chair. The positions shall be accomplished by operating a seat control lever located under the seat. The chair shall have the capability for mounting auxiliary accessories such as a light or tray bracket post by means of an adaptor mounted on the top of the chair pivot casting. Attached to the seat section shall be the toe section which shall support the patient’s feet and lower legs. The toe section shall be easily removable for storing In the case. The toe section shall also have a clear plastic cover to protect the toe section upholstery from dirt and scuffs. Attached to the rear of the seat section on pivot bolts shall be the back section which shall provide a means of patient back and head support through all positions of the back travel. Back angular adjustment shall be accomplished by depressing a button (back release control) located at the top of the back which shall operate a spring actuated hydraulic mechanism. The mechanism shall act as an assist when raising the patient from a horizontal position and shall provide the locking means to lock the back in any position through its full angular travel. The patient’s arms shall be supported by slings (arm supports) mounted to the upper side of the back and shall be capable of being attached to and removed from spring loaded arms mounted on seat frame. The chair shall have a removable and adjustable “U” shaped, cushioned, vinyl upholstered headrest. The color of the chair shall be as specified in 3.6.1.

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3.4.3 Stool (See figure 1). The stool shall be a mobile, adjustable dental operating stool designed with an anatomical (saddle type) seat for use during dental operating procedures. The stool shall be sturdy and stable, and shall provide proper support for body weight during repeated use. The stool shall have a cushioned, vinyl upholstered seat that shall be adjustable in height, and a base assembly consisting of a cross type base casting with casters and a foot ring. The seat shall be mounted on a tapered roller bearing for ease of swiveling. The stool shall have a height adjustment shaft which slides within a supporting post and shall have notches to receive a locking plunger (stop assembly) to permit adjustable seat height. The stool shall be capable of being disassembled for storage in the case. The stool shall meet the requirements of 3.5.2. The color of the stool shall be as specified in 3.6.1.

3.4.4 Case. The carrying case shall fully envelop the field chair and stool in the module's collapsed, bolted and strapped storage position (See figure 4). The carrying case shall be water-vaporproof (Type II, Style 2, of MIL-C-4150) and shall meet the applicable requirements of MIL-C-4150 as modified by the applicable drawings. The color of the case shall be as specified in 3.6.2. The marking and nameplate requirements shall be as specified in 3.12.

3.5 Range of travel and angular adjustment.

3.5.1 Chair. The chair in its set up operating position shall meet the adjustable ranges specified in figure 6.

3.5.2 Stool. The stool shall be adjustable from a low position of 18 ± 1 inches (top of the seat above the floor) to a high position of 25 ± 1 inches (top of the seat above the floor) in seven increments when tested as specified in 4.5.8.2.

3.6 Color.

3.6.1 Chair and stool. The chair upholstery, the chair base cover, the stool upholstery and the thread used for stitching the upholstery shall be beige (see 3.16 and 6.1).

3.6.2 Carrying case. The carrying case color shall be No. 34087 - Olive Drab (lusterless) in accordance with FED-STD-595 (See 3.16 and 6.1).

3.7 Caution marking. Caution marking on the chair with regard to positioning of the chair when removing the chair from the" case or repositioning the chair shall be in accordance with the applicable drawing and shall be placed on the front face of the base cover and on the back of the chair as shown on the applicable drawing. The case shall be marked with arrows as specified in paragraph 5.2.2.2 of MIL-STD-129. The carrying case relief valve shall have a caution label placed adjacent to the valve on the outside of the case. The relief valve label shall be an anodized aluminum label having a red background. The relief valve caution label shall read:

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RELIEF VALVE
CAUTION
UNSCREW BEFORE
UNLOCKING CASE
SCREW TIGHTLY
AFTER LOCKING CASE

All labels shall be secured with a permanent pressure sensitive adhesive.

3.8 Weight. The weight of the module and major components shall be as follows when weighed in accordance with 4.5.2:

Complete chair including floor plate	110 ± 5 pounds
stool	18 ± 1 pounds
Carrying case	44 ± 1 pounds
Complete module (chair, stool, and case)	172 ± 7 pounds

3.9 Finish. All surfaces shall be clean and free of foreign matter, flux or spatters of weld. The finish of components shall be as specified on the applicable drawing or list, and as specified herein. If no finish is specified, the finish shall be entirely suitable for the purpose intended for the applicable component while providing necessary protection from deteriorating effects of climate and service conditions encountered in military uses. Finish on edges and surfaces shall be uniform and free from burrs, slivers, sharp edges, cracks, or other defects.

3.9.1 Plating (coatings). Plating and coatings shall be as specified on the applicable drawing or list, and as specified herein. When no thickness of plating is specified, the plating shall be of sufficient thickness to be suitable for the purpose intended for the applicable component. plating shall be uniform in thickness and appearance, shall be free from blisters, peeling, visible porosity, or other defects. The base metal to be plated shall be free from flaws or defects which will be detrimental to the final finish and shall be subjected to such undercoating, polishing, cleaning, dipping and plating procedures as are necessary to yield deposits with the desired function, quality and appearance.

3.9.2 Upholstery. Upholstery and upholstery texture shall be as specified on the applicable drawing or list, and as specified herein.

3.10 Workmanship. Workmanship shall be first class throughout and in accordance with best commercial practice. All components shall be clean, well finished, and free from defects which detract from their appearance or may impair their serviceability.

3.10.1 Upholstery. The upholstery shall be neatly cut, sewed, and sealed, where specified. The upholstery shall neatly and uniformly surround the particular component with sufficient tension to maintain a smooth appearance before and after use. The upholstery shall not contain any cuts, tears, rips, punctures, or any defects which shall impair serviceability or detract from appearance.

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3.10.2 Stitching and seams. Workmanship shall be first class and shall be in accordance with best commercial practice. All thread and stitching , shall be as specified on the applicable drawing, and as specified herein. Seam allowances shall be maintained with seams properly sewn so that no raw edges, runoffs, twists, pleats , puckers or open seams shall result. There shall be no thread breaks or skipped stitches. Unless otherwise specified, all seams shall be straight and evenly spaced.

3.10.3 Heat seals. Heat seals shall be neat, uniform and provide a sound and strong seam. The heat seal seam shall not be blistered, cracked, creased, or wrinkled. The seam shall be continuous and provide a complete secure seam throughout the entire seam area.

3.10.4 Castings. Castings shall be sound and free from repairs, warping, or other defects that may render the casting unsound for use. Repair processes, such as welding, peening, plugging, or filling with cold solder or metallic paste shall not be permitted.

3.10.5 Welding, brazing and soldering. All welds, brazing, and soldering shall be durable and sound in every detail. Welding shall be in accordance with the latest practices recommended by the American Welding Society. Components shall be joined without any resultant distortion and shall be free from twists or strains. All welds shall be produced by qualified welding operators with satisfactory welding equipment using satisfactory filler metal. Surfaces to be welded shall be free from grease, oil, oxides or any other foreign matter. Any cleaning method which will remove these substances and leave the surface clean is satisfactory. All aluminum parts shall be welded within 48 hours. after a chemical cleaning treatment. Parts must be chemically recleaned if the 48 hour time period is exceeded. All weld joints shall blend into the adjacent metal in gradual, smooth curves. Beads shall be smooth and free of slag, excessive undercut, or excessive spatter. Sufficient metal shall be added to provide a suitable fillet or reinforcement unless otherwise specified. Welds shall not be oxidized. All welds shall be continuous with no visible voids, pin holes, or cracks. Exposed welds shall be smooth and blend with adjacent surfaces. All welds shall be cleaned, discoloration removed, and be free from slag and spatter before finishing. Welds shall be ground, only to the degree necessary to remove sharp edges or projections which may cause defects to adjacent surfaces or can be injurious to the touch.

3.10.6 Machine work. Machining shall be first class and in accordance with best commercial practice. All bolt, machine screw and rivet holes, shall be accurate and clean. If not specified, the holes for screws and bolts shall be sized appropriately for the size screw or bolt used. All bolts, nuts, screws, and fasteners shall be completely fastened and tight. When tolerances are not specified for machining operations, tolerances shall be such as to be entirely suitable for the purpose intended.

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3.10.7 Metal fabrication. Metals (sheets, plates and tubing) shall be free from dents, folds, or distortions. Material shall be straightened by methods that will not cause injury to the metal. Operations (such as bending, shearing, or cutting) shall be done neatly and accurately. Metal forming shall be made with controlled means to insure uniformity of size and shape.

3.11 Hardware. Hardware shall be as specified in the applicable drawing, list, and as specified herein. If not specified, all hardware shall be entirely suitable for the purpose intended and shall be completely compatible with the material to which it is attached or which it joins including plated material. Any fastening device required to be loosened, tightened, or removed for repair, maintenance or adjustment shall not be swaged, peened, staked, or otherwise deformed to prevent loosening of the device. All rotative fasteners shall be correctly torqued. All threaded parts shall comply with the National Bureau of Standards Handbook H-28.

3.12 Identification marking. Each module component (chair, stool and case) shall be marked in accordance with the applicable requirements of MIL-STD-130 as modified by this specification. Each case shall be marked with the following information:

6520-00-181-7349
CHAIR AND STOOL UNIT, DENTAL OPERATING, PORTABLE
Name or registered trademark of the manufacturer
Contract or Purchase Order Number
Serial Number
Letters "US"

Each chair shall be marked with the following information:

Chair Unit of 6520-00-181-7349
Name or registered trademark of the manufacturer
Contract or Purchase Order Number
Serial Number
Letters “US”

Each stool shall be marked with the following information:

Stool Unit of 6520-00-181-7349
Name or registered trademark of the manufacturer
Contract or Purchase Order Number
Serial Number
Letters "US"

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The marking of the case shall be located on the outside top face of the carrying case. The marking of the chair shall be located on front face of the lift arm (see figure 5), as specified on the applicable drawing. The marking of the stool shall be located on the support post as specified on the applicable drawing. In addition to the above marking, each case (top and bottom portion) shall be marked with the serial number of the case on the mating metal extruded portions of the case such that the numbers can be matched to identify the mating portions of the case.

3.13 Tolerances. Standard commercial tolerances shall be permissible on material gage sizes. Dimensional tolerances shall be as specified on the drawings. Tolerances shall be held to a minimum where accuracy is of paramount importance for proper operation and interchangeability of component parts. Assemblies shall be precision-fitted for durable, smooth, and consistent functioning in keeping with first class workmanship throughout.

3.14 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.15 Hydraulic charge. Prior to shipping the module, the supplier shall charge the hydraulic system so that the chair shall be in the ready to use condition.

3.16 Approval of colors and cushioning. Samples of acceptable upholstery color and grain texture, base cover color, thread color, and chair cushioning material are available for inspection at DPSC. Samples of the upholstery for color and grain texture, base cover color and cushioning material (canvas backed foam, chair seat and stool padding) shall be submitted to the contracting officer for approval prior to commencing production. (See 6.1).

3.17 Tests. (See definitions 6.2.1 through 6.2.5).

3.17.1 Chair tilt and deformation.

3.17.1.1 Chair tilt and deformation (weighted back section, unloaded chair.) The assembled chair shall not tilt (upon application of weights), and there shall be no deformation (after weights are removed) when tested as specified in 4.5.3.1.

3.17.1.2 Chair tilt and deformation (weighted back section, loaded chair). The assembled chair shall not tilt (upon application of weights and load), and there shall be no deformation (after weights and loads are removed) when tested as specified in 4.5.3.2.

3.17.1.3 Chair tilt and deformation (weighted toe section, unloaded chair). The assembled chair shall not tilt or deflect (more than 3-1/2 inches) at the toe section with weight applied, and there shall be no deformation (after weight is removed) when tested as specified in 4.5.3.3.

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3.17.2 Chair stability. The assembled chair shall remain stable, shall not 'tilt, and shall not deflect more than the amount specified for each of the positions shown in figures 10 through 13 upon application of a 300 pound distributed load when tested as specified in 4.5.4. The chair shall show no deformation after removal of the weights in each position.

3.17.3 Arm operation.

3.17.3.1 Arm spring force. The force required to hold the arm at its rearmost position shall be 15 ± 2 pounds when tested in accordance with 4.5.5.1.

3.17.3.2 Arm spring operation. The arm and control box assembly shall show no signs of failure, sideplay, deformation or other defects when tested as specified in 4.5.5.2.

3.17.4 Hydraulic system performance. After assembly of the hydraulic system, not assembled in the chair. The hydraulic system shall show no signs of malfunctioning (see 6.2.5) under loads of 350 ± 5 pounds for a minimum of 30 minutes and 700 ± 5 pounds for a minimum of 2 minutes when tested as specified in 4.5.6.

3.17.5 Overall chair operation.

3.17.5.1 Operation of the chair (linear and angular movement). The assembled chair shall be capable of moving through the positions shown, and shall meet the linear and angular dimensions within the respective positions established in figure 6, when tested as specified in 4.5.7.1.

3.17.5.2 Operation of the chair (unloaded). The assembled chair (all operating components, adjustments and controls) shall operate quietly, smoothly, efficiently and shall show no signs of defects or malfunctioning when tested as specified in 4.5.7.2.

3.17.5.3 Operation of the chair (loaded). The maximum number of complete operations of the foot lever to raise the chair from its lowest to its highest position under the loads specified shall be as specified in table I when tested as specified in 4.5.7.3.

Table I. Maximum operations of foot lever under various loads.

Load (pounds)	Maximum operations of the foot lever
No load	22
150 +0, -3	23
200 +0, -4	25
300 +0, -6	29
450 +0, -8	33

The chair shall show no evidence of failure after each test.

3.17.6 Stool operation. The assembled stool shall securely lock in each increment (each notch of the adjustment shaft) and shall show no signs of malfunctioning or defects when operated from its lowest position to its highest position as specified in 3.5.2 when tested in accordance-with 4.5.8.

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3.17.6.1 Stool loading. The assembled stool shall withstand a minimum load of 500 pounds for a minimum of two weeks without any resultant malfunctioning, defects, failures, deformation (including permanent set and creasing, or other defects, in the cushioning material or upholstery) when tested as specified in 4.5.8.1. (See 6.2.4 and 6.2.5).

3.17.7 Module assembly and disassembly.

3.17.7.1 Module assembly. The chair and stool shall be capable of being assembled and set up using the assembly procedures specified in 4.5.9 (see figure 1 for assembled chair and stool illustration).

3.17.7.2 Module packaging (in carrying case). The chair and stool shall be capable of being disassembled, packaged and secured in the carrying case. The packaging procedures shall be as specified in 4.5.10 (see figure 3 for chair collapsed position illustration and figure 4 for module component orientation).

3.18 Service Data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the Contracting Officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. In accordance with 4.1, the supplier shall insure that components and materials used are manufactured, examined, and tested in accordance with referenced drawings, specifications, and standards unless excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.1.2 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.4 Certificates of quality. Certificates of quality, supplied by the manufacturer of materials and components, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the contractor, the contract number, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements. Certificates of quality may be furnished for the following: material composition, hardness, tempers, torque requirements on springs, machining and casting requirements (such as flatness, concentricity, and finish), and plating or coating.

4.1.5 Certificate of Compliance. The carrying case may be accepted on the basis of a contractor's certification of compliance with the applicable requirements of MIL-C-4150 as modified by the applicable drawings. Color and marking requirements shall be as required by this specification. The Government reserves the right to check test any item to determine the validity of any certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).
3. Preparation of delivery (see 4.4.6).

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4.3 First article inspection. When a first article is specified in the contract, the first article inspection shall consist of all examinations specified in 4.4.1.1, 4.4.2.1, 4.4.3, and all testing specified in 4.4.4, 4.4.5, and shall apply to the first article sample only (see 6.1 and 6.3). The failure of compliance with any examination or test shall be cause for rejection of the first article. Prior to approval of the first article, if a first article is "required, or prior to commencing production the supplier shall submit to the contracting officer satisfactory evidence that the carrying case he proposes to furnish under this specification meets the applicable requirement of MIL-C-4150, as modified by the applicable drawings.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 In-process examination. In-process examination shall be conducted to ascertain that components or assemblies which cannot be readily examined in the end item are in accordance with specification requirements.

4.4.1.1 Visual examination. In-process examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table II. The inspection level shall be S-3 with an acceptable quality level (AQL) of 2.5 for major defects and 40 for total defects, expressed in terms of defects per hundred units. The unit of product shall be the component of an end item or an end item, as applicable.

Table II. In-Process Classification of defects

Welding, Brazing & Soldering	Major	Minor
Missing, incomplete, fractured, not fused, not continuous, not sound.	x	
Burn holes, pits.		x
Rough and not uniform.		x
Not free of scale, rust, flux, grease, foreign matter.		x
Welds have sharp edges or projections which may cause holes, tears, impair the upholstery or cause injury when touched.		
Excessive solder on joint (runoff).		x

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Table II. In-Process Classification of defects (cont'd)

Hardware	Major	Minor
Threads damaged.		x
Finish		
Finish not as specified.	x	
Surfaces not clean, not free o-f foreign matter, flux, spatters, or other defects.		x
Damage or defects on exterior or interior surfaces such as burrs, slivers, cracks, sharp edges, which affect function or serviceability.	x	
Damage or defects on exterior or interior surfaces such as slivers, cracks, sharp edges which do not affect function or serviceability.		x
Plating or coating missing (affecting function).	x	
Plating or coating missing (not affecting function).		x
Plating not uniform in appearance.		x
Plating not free from blisters, peeling, visible porosity, or other defects (affecting function or serviceability).	x	
Plating not free from blisters, peeling, visible porosity, or other defects (not affecting function or serviceability).		x
Workmanship		
Any component fractured, broken, punctured, torn, bowed, deteriorated, or malformed.		
Any component dented or creased.		x
Any component not connected or not joined properly.		x
Fastening device requiring loosening or removal is swaged, peened, staked or otherwise permanently fastened.	x	
Components missing.	x	

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Table II. In-Process Classification of defects (cont'd)

Workmanship (cont'd)	Major	Minor
Interface fits between components not proper (too loose, too tight, binding).	X	
Components not free from defects which may impair their serviceability.	X	
Components not free from defects which do not impair their serviceability.		X
Castings		
Cast components not sound, not free from porosity, cracks or blowholes.	X	
Cast components repaired, or brazed.	X	
Cast components malformed.	X	
Machine Work		
Holes not accurately and cleanly made.	X	
Coarse machine tool or die marks which affect function or serviceability.	X	
Coarse machine, tool or die marks which do not affect function or serviceability.		X
Components not accurately and smoothly machined on contacting surfaces.	X	
Heat Seals		
Seam not neat and uniform.		X
Seam blistered, cracked, or creased.		X
Seam not continuous or secure.	X	
Seam open.	X	

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Table II. In-Process Classification of defects (cont'd)

Metal Fabrication (Sheets, plates and tubing)	Major	Minor
Metal not free from dents, folds, or distortions (seriously affecting appearance or serviceability).	x	
Metal not free from dents, folds or distortions (not seriously affecting appearance or serviceability) .		x
Shearing, bending, or cutting not neat and accurate "(seriously affecting appearance, serviceability, or can cause injury).	x	
Shearing, bending, or cutting not neat and accurate (not seriously affecting appearance, serviceability or cannot cause injury).		x
Forming not neatly finished.		x

4.4.2 End item examination.

4.4.2.1 Visual examination. End item examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table III. The inspection level shall be S-3 with an acceptable quality level (AQL) of 2.5 for major defects and 40 for total defects, expressed in terms of defects per hundred units. The unit of product shall be the component of-an end item or an end item, as applicable.

Table III. End Item Classification of defects

Design and Construction	Major	Minor
Design and construction of chair not as specified.	x	
Design and construction of stool not as specified.	x	
Design and construction of case not as specified.	x	
Color		
Color of upholstery not as specified.	x	
Color of case not as specified.	x	
Color of base cover not as specified.	x	

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Table III. End Item Classification of defects (cont'd)

Cushioning material	Major	Minor
Not flexible.	X	
Not resilient.	X	
Hardware		
Components such as pins, screws and fasteners missing, broken, stripped, cracked, or deformed.	X	
Fasteners loose.		X
Finish		
Finish not as specified.	X	
Surfaces not clean, not free of foreign matter, flux, spatters, or other defects.		X
Damage or defects on exterior or interior surfaces such as burrs, slivers, cracks, sharp edges, which affect function or serviceability.	X	
Damage or defects on exterior or interior surfaces such as slivers, cracks, sharp edges which do not affect function or serviceability.		X
Plating or coating missing (affecting function).	X	
Plating or coating missing (not affecting function).		X
Plating not uniform in appearance.		X
Plating not free from blisters, peeling, visible porosity or other defects (affecting function or serviceability) .	X	
Plating not free from blisters, peeling, visible porosity or other defects (not affecting function or serviceability).		X

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Table III. End Item Classification defects. (cont'd)

Workmanship	Major	Minor
Any component fractured, broken, punctured, torn, bowed, deteriorated, or malformed.	X	
Any component dented or creased.		X
Any component misplaced or not in proper alignment.	X	
Any component not connected or not joined properly.		X
Fastening device requiring loosening or removal is swaged, peened, staked or otherwise permanently fastened.	X	
Components missing.	X	
Components do not fit or mate properly.	X	
Hydraulic system not charged with hydraulic fluid.		X
Interface fits between components not proper (too loose, too tight, binding).	X	
Components not free from defects which may impair their serviceability. "	X	
Components not free from defects which do not impair their serviceability.		X
Removable components (such as toe section) cannot be removed and replaced without difficulty.	X	
Components not properly assembled or aligned.	X	
Chair or stool not stable on floor (not level, contains rock motion).	X	
Stool adjustment shaft does not operate smoothly.	X	
Casters on stool do not operate or swivel freely or bind.		X
Stool seat does not operate or swivel freely or bind.	X	

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Table III. End Item Classification defects (cont'd)

Machine Work	Major	Minor
Coarse machine tool or die marks which affect function or serviceability.	X	
Coarse machine, tool or die marks which do not affect function or serviceability.		X
Upholstery		
Permanently stained or spotted in a way that is clearly noticeable and that cannot be removed with a cleaning agent.		
Stained or spotted (not permanently) in a way that is noticeable but that can be easily removed with a cleaning agent.		X
Color not uniform seriously affecting appearance.	X	
Color not uniform, but not seriously affecting appearance.		X
Grain not as specified.	X	
Grain indistinct or not uniform.		X
Upholstery cut, ripped, punctured.	X	
Contains open scratches or abrasions having a serious effect on serviceability or appearance.	X	
Contains scratches or abrasions not having a serious effect on serviceability or appearance.		X
Contains creases, or wrinkles seriously affecting serviceability or appearance.	X	
Contains creases, or wrinkles not seriously affecting serviceability or appearance.		X

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Table III. End Item Classification defects (cont'd)

Upholstery (cont'd)	Major	Minor
Upholstery not same on any one component or does not match any other component of the module.	X	
Upholstery not free of objectionable odor*	X	
Upholstery patched or mended (seriously affecting serviceability or appearance).	X	
Upholstery patched or mended (not seriously affecting serviceability or appearance).		X
Heat Seals		
Seam blistered, cracked, or creased.		X
Seam not continuous or secure.	X	
Seam open.	X	
Stitching		
Loose or broken stitching resulting in poorly secured seam.	X	
Tight stitching resulting in rupture of upholstery.	X	
Open seams caused by one or more broken stitches, or two or more consecutive skipped stitches or runoffs which result in a poorly secured seam.	X	
Open seams caused by one or more broken stitches, or two or more consecutive skipped stitches or runoffs which do not result in a poorly secured seam.		X
Stitching type not as specified.	X	
Seams not straight or evenly spaced.		X

*Aromatic odors usually associated with specific types of material shall not be regarded as objectionable.

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Table III. End Item Classification defects (cont'd)

Stitching (cont'd)	Major	Minor
Seam pleated, puckered badly, or twisted.	X	
Stitch count less than 4 stitches per inch.	X	
Stitch count greater than 8 stitches per inch.	X	
Stitch count 8 or less, but greater than 6 stitches per inch.		X
Needle chews (affecting appearance seriously).	X	
Needle chews (not affecting appearance seriously).		X
Raw edge of material noticeable at seam.		X
Thread ends not trimmed off neatly (seriously affecting appearance).	X	
Thread ends not trimmed off neatly (not seriously affecting appearance).		X
Color of thread not as specified.		X
Damaged stitching or seam after installation (end item) which seriously affects appearance or serviceability.	X	
Damaged stitching or seam after installation (end item) which does not seriously affect appearance or serviceability.		X

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Table III. End Item Classification defects (cont'd)

Identification Marking	Major	Minor
Identification marking missing, incomplete, not legible.		X
Identification marking incorrect or not permanent.		X
Caution plates missing, incomplete, not legible.		X
Caution plates not permanently secured.		X
Marking not properly located.		X
Service Data		
Service data not included.	X	

4.4.3 Dimensional examination. Dimensional examination shall be conducted for compliance with dimensions specified. Any dimension not within specification tolerance shall be a defect. The inspection level shall be S-3 with an AQL of 10 expressed in terms of defects per hundred units. Unit of product shall be the component of an end item or an end item, as applicable.

4.4.4 In-process testing. In-process testing shall be conducted to ascertain that components or assemblies which cannot be readily tested in the end item are in accordance with specification requirements. In-process testing shall be conducted in accordance with MIL-STD-105 and as indicated in table IV, except that the acceptance number shall be zero for all sample sizes. A failure during 100 percent inspection shall cause rejection of that component or item only (not the entire lot). Unit of product shall be as specified in table IV.

4.4.5 End item testing. End item testing shall be conducted in accordance with MIL-STD-105 and as indicated in table IV, except that the acceptance number shall be zero for all sample sizes. A failure during 100 percent inspection shall cause rejection of that component or item only (not the entire lot). Unit of product shall be as specified in table IV.

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Table IV. Sampling and Acceptance Criteria for Testing

Characteristic	Requirement	Test Procedure	Inspection Level	Unit of Product
Composition Analysis	3.1	4.5.1	See 4.4.5.1	-
In Process Testing				
Arm spring force 1/	3.17.3.1	4.5.5.1	100%	One arm assembly
Arm spring operation 1/	3.17.3.2	4.5.5.2	S-4	One arm assembly
Hydraulic system performance testing.	3.17.4	4.5.6	100%	One hydraulic system
End item testing				
Weight	3.8	4.5.2	2/	One module
Tilt and deformation weighted back, unloaded chair.	3.17.1.1	4.5.3.1	2/	One chair
weighted back, loaded chair.	3.17.1.2	4.5.3.2	2/	One chair
weighted toe section, unloaded chair.	3.17.1.3	4.5.3.3	2/	One chair
Chair stability test.	3.17.2	4.5.4	2/	One chair
Operation of chair (linear and angular movement).	3.17.5.1	4.5.7.1	2/	One chair
Operation of chair (unloaded)	3.17.5.2	4.5.7.2	100%	One chair
Operation of chair (loaded)	3.17.5.3	4.5.7.3	S-2	One chair
Stool operation	3.17.6	4.5.8	100%	One stool
Stool loading	3.17.6.1	4.5.8.1	2/	One stool
Stool adjustable height	3.5.2	4.5.8.2	S - 2	One stool
Assembly test	3.17.7.l	4.5.9	2/	One module
Packaging test	3.17.7.2	4.5.10	2/	One module
Arm spring force 1/	3.17.3.1	4.5.5.1	100%	One chair
Arm spring operation 1/	3.17.3.2	4.5.5,2	S-4	One chair

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1/ This test may be conducted in process or on the assembled chair (end item).

2/ Three samples representative of each lot shall be selected for test.

4.4.5.1 Raw materials. Sufficient amount from each lot of each material used in the construction of the module shall be selected for chemical analysis.

4.4.6 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing and marking complies with section 5 requirements. The sample unit shall be one shipping container and the lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the acceptable quality level (AQL) shall be 4.0 defects per hundred units.

Examine	Defect
Markings (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location sequence or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as: Incomplete closure of container, loose or inadequate sealing, strapping or stapling. Wrap or bag not securely closed. Bulged or distorted container.
Content	Not as specified.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Composition analysis. Chemical analysis of metal materials used shall be subjected to analysis in accordance with methods 111 or 112 Federal Test Method-Standard No. 151. In case of dispute, method 111 shall be used for referee analysis. Chemical analysis of materials other than metal shall be determined by an applicable American Society for Testing and Materials (ASTM) Method.

4.5.2 Weight. The chair, stool, case and complete module shall be weighed on a calibrated scale of suitable sensitivity and accuracy.

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4.5.3 Chair tilt and deformation.

4.5.3.1 Chair tilt and deformation (weighted back section, unloaded chair). The chair shall be mounted in the bottom portion of the carrying case. The chair shall be positioned and weighted as shown in figure 7 for a minimum of 3 minutes. Upon application of weights, the chair shall be observed for tilt. After removal of weights, the chair shall be examined for deformation.

4.5.3.2 Chair tilt and deformation (weighted back section, loaded chair). The chair shall be mounted in the bottom portion of the carrying case. The chair shall be positioned, weighted and loaded as shown in figure 8 for a minimum of 3 minutes. Upon application of weights and loads, the chair shall be observed for tilt. After removal of weights and loads, the chair shall be examined for deformation.

4.5.3.3 Chair tilt, deflection and deformation (weighted toe section, unloaded chair). The chair shall be positioned and weighted at the toe section as shown in figure 9 for a minimum of 1 minute. Upon application of weight, the chair shall be observed for tilt. With weight applied, deflection shall be measured at the free end of the toe section. After removal of weight, the chair shall be examined for deformation.

4.5.4 Chair stability. The chair shall be mounted in the bottom of the carrying case. The chair shall be positioned and weighted as shown in figure 10. With weights applied, the chair shall be observed for tilt and measured for deflection at the back and toe section. Weights shall be removed and the chair shall be examined for deformation. The entire procedure shall be repeated with the chair positioned and weighted as shown in figures 11 through 13.

4.5.5 Arm Operation.

4.5.5.1 Arm spring force. The arm control box (completely assembled) shall be mounted in a vise. As an alternate, the arm spring force can be tested on the assembled chair. The arm shall be pulled back using suitable instrumentation such as a spring scale and the force required to hold the arm in its rearmost position shall be noted. The force shall be tangent to the free end section of the arm.

4.5.5.2 Arm spring operation. With the control box mounted in a vise or on the end item as described in 4.5.5.1, the arm shall be operated through 50 complete cycles (one cycle shall be a complete back and forth movement). The arm assembly shall then be examined for failures or defects.

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4.5.6 Hydraulic system performance. (After assembly of the hydraulic system, not assembled in the chair). After assembly of the hydraulic system, with the system properly filled with fluid and purged (air removed), the hydraulic system shall be subjected to a load of 350 ± 5 pounds for a minimum of 30 minutes and 700 ± 5 pounds for a minimum of 2 minutes. The load shall be applied by utilizing a fixture containing a spring (calibrated for the specified loads) such that the cylinder rod of the cylinder assembly can be pumped against the spring in a vertical position until the specified load is applied to the system. The system shall be observed during the test and inspected after each load application.

4.5.7 Overall chair operation.

4.5.7.1 Operation of the chair (Linear and angular movement). The assembled chair shall be operated to the positions shown in figure 6 and the linear and angular dimensions shall be measured.

4.5.7.2 Operation of the chair (unloaded). The chair shall be subjected to operational testing which shall consist of, but shall not be limited to, the following systems or components. Each system or component shall be operated for a minimum of three operations or cycles (as applicable).

Hydraulic system operation (raising and lowering of the chair, operation of speed control valve).

Back position control operation (functional including capability of locking the back in any position through entire range of travel).

Operation of seat pivot (“UP and “DOWN” position) see figure 6.

Operation of release lever (with chair attached to the case bottom).

Operation of arm assemblies.

Toe section (attachment and removability).

Head rest (attachment and adjustability).

4.5.7.3 Operation of the chair (loaded). The number of pumps (operations of the foot lever) to raise the chair from its lowest to its highest position under loads as specified in 3.17.5.3 shall be determined. The distribution of loads shall be as specified in figures 14 through 17. The chair shall be examined for defects such as leakage after each test.

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4.5.8 Stool operation. The stool shall be operated and locked in each position of the shaft (each notch) and examined for correct interface fit. , The stool shall meet the requirements of 3.17.6.

4.5.8.1 Stool loading. The fully assembled stool shall be raised to its highest position and a minimum load of 500 pounds shall be placed on the seat and allowed to remain for a minimum of 2 weeks. The load shall consist of sand bags placed on a flat, square member dimensionally greater than the greatest diameter of the stool seat. After this time, the load shall be removed and the stool shall be inspected and operated. The stool shall meet the requirements of 3.17.6.1.

4.5.8.2 Stool adjustable height. The adjustable height of the stool shall be measured when locked in its lowest and highest position.

4.5.9 Module assembly. The chair and stool shall be assembled using the following sequence. The sequence starts with the entire module packaged in the carrying case.

1. Remove spring clamps on exterior portion of carrying case and remove top portion.
2. Remove packaging strap that holds dentist’s saddle seat and stool base (and remove base).
3. Remove toe section and install on lower section of the chair seat. Snap tabs on bottom of toe section to seat section.
4. Remove connecting pin from power actuator and allow safety latch to drop down.
5. Lift back section in upright position and engage connecting pin through bracket and piston assembly. Pull back on each spring loaded arm and engage arm supports. (Stool supporting post and chair foot lever extension are packaged on seat section).
6. Loosen wing nuts on stool base and mount stool post in opening provided. Tighten wing nuts to secure post. Pull out spring loaded knob on stool supporting post and install saddle seat.
7. Disengage seat control lever and move chair seat upward (toe section horizontal) and engage seat control lever.
8. Install extension pedal on foot lever.

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4.5,10 Module packaging. The module shall be packaged using the following sequence. The sequence starts with the chair and stool fully assembled.

1. Lower chair base to full "DOWN" position.
2. Unsnap tabs beneath toe section securing toe section to seat section and remove.
3. Disengage seat control lever and position seat downward (toe section down) . Engage seat control lever.
4. Bring chair back to approximately 70 degree angle. Remove connecting pin. Attach safety latch with connecting pin to chair actuator. CAUTION: Be sure this is accomplished before proceeding so that dentist's saddle seat will properly fit. If back release control button is accidently depressed before the safety latch is secured it will be necessary to re-assemble chair back to power actuator to acquire the proper position of the actuator in relationship to the safety latch.
5. Disengage arm supports, fold and place on chair seat. Flip up foot operated release lever against end shield.
6. Remove stool seat together with adjustment shaft. Loosen wing nuts on stool base and place stool post (in cushioned bag) on chair seat. Tighten wing nuts.
7. Place saddle seat in front of chair base.
8. Place stool chassis in rear area with casters facing chair.
9. Place packaging strap around stool seat, chair, and stool base and secure.
10. With stool post on chair seat, place chair back down on chair seat and toe section on top of chair back.
11. The carrying case bottom portion has a serial number on the aluminum extrusion. The carrying case top portion has the same serial number. Place the top portion on the bottom portion so that the serial numbers match and secure spring loaded clamps.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit-of issue. One portable operating chair and stool in carrying case, as specified, constitutes one unit of issue.

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5.1.2 Level A. Each chair and stool shall be packaged within the carrying case as specified in 4.5.10. The stool supporting post and adjustment shaft shall be adequately wrapped and cushioned prior to positioning in the carrying case. The carrying case shall be inserted into a 3 mil plastic bag. Excess air shall be evacuated from the bag and the closure effected by heat sealing. As an alternate, the case may be wrapped in waterproof barrier material conforming to MIL-L-10547. Closure and sealing shall be as specified in the appendix of MIL-L-10547.

5.1.3 Level C. Each unit shall be packaged as specified in 5.1.2 except that the carrying case shall be packaged in a standard commercial container of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A. Each unit packaged as specified in 5.1.2 shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-601, overseas type or PPP-B-640, class 2, style E. Container shall include suitable interior fittings to protect exterior hardware and carrying handles on the carrying case and to prevent movement during shipping and handling. Closure and strapping shall be as specified in the appendix of the applicable box specification except that the use of steel strapping is not authorized for fiberboard boxes.

NOTE: Strapping shall not be required on wood boxes for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level C. The subject commodity shall be packed in substantial commercial containers (export when applicable) of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking. Exterior container shall be marked in accordance with MIL-STD-129. In addition, marking shall include arrows as specified in 5.4.24 of MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Approval of substitution of proprietary components (see 3.3).
(c) When a first article is required (see 3.3, 4.3 and 6.3).
(d) Approval of colors and cushioning (see 3.6 and 3.16).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Definitions. The term (see 3.17) as used in this specification shall be defined as specified herein.

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6.2.1 Normal sitting position. The normal sitting position of the chair (see figures 9 and 13 through 17) shall be when the toe section is horizontal, with the seat in the "UP" position (hole A of the seat adjustment support, figure 6), and the back in its most forward position (position 3, figure 6).

6.2.2 Sand bags. A sand bag (see figures 9 through 17) shall consist of a bag contained with sand and weighing 50 ± 1 pounds. The approximate dimensions of the loaded (filled) portion shall be 17 inches in length, 14-1/2 inches in width and 6 inches in height.

6.2.3 Tilt. Tilt (see 3.17.1) as used in this specification shall be any tipping or movement which causes any separation between the floor and the chair floor plate (or case bottom if the chair is mounted in the case bottom during the test).

6.2.4 Deformation. Deformation (see 3.17.1) shall be a change in structure of any member (such as a permanent set, bending, twisting, fracturing, or rupturing). Deformation shall be determined by visual means (perceptible to the naked eye), with the aid of suitable instrumentation such as rulers, straight edges or protractors. Instrumentation for measuring linear measurements shall have 1/16 inch or finer graduations and for measuring angular measurements 1/2 degree or finer. Unless otherwise specified, to determine deformation linear or angular measurements shall be measured to the nearest 1/16 inch or 1/2 degree.

6.2.5 Malfunctioning. Malfunctioning (see 3.17.4, 3.17.5.2, 3.17.6, and 3.17.6.1) shall be any evidence that a component, assembly or entire item is not properly or effectively operating or any evidence of an existent or potential defect which may cause improper or ineffective operation or failure (such as binding, sticking, leakage, poor interface fit, rubbing between mating parts, non-locking of locking components, excessive play, bending, excessive force required to operate components, loss of pressure, or other defects).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

Stock No.	Item Identification
6520-00-181-7349	CHAIR AND STOOL UNIT, DENTAL OPERATING, Portable: Field type. In carrying case.

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6.5 Limitations. This specification does not cover all types, classes, grades, styles, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

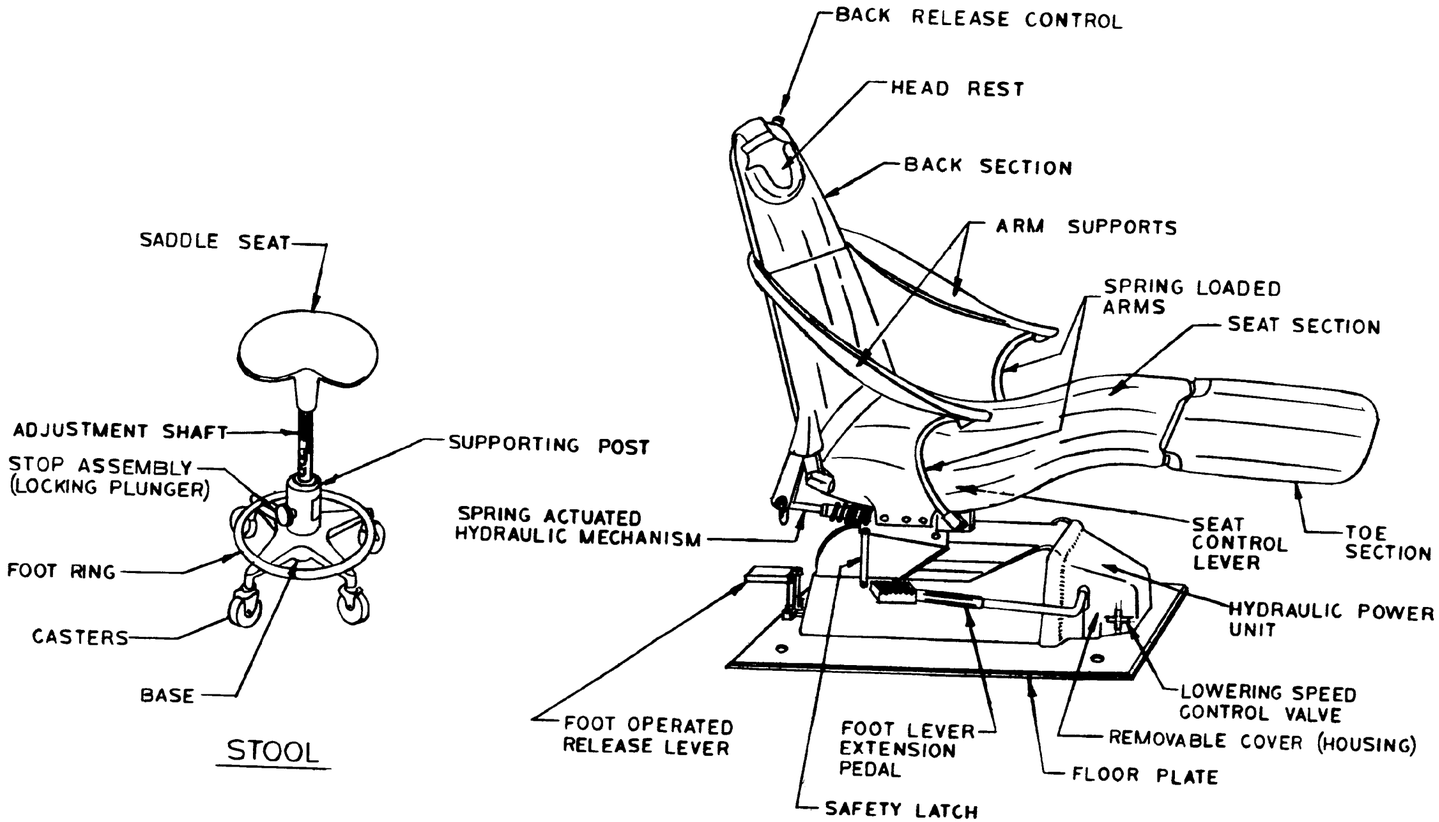
Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1366

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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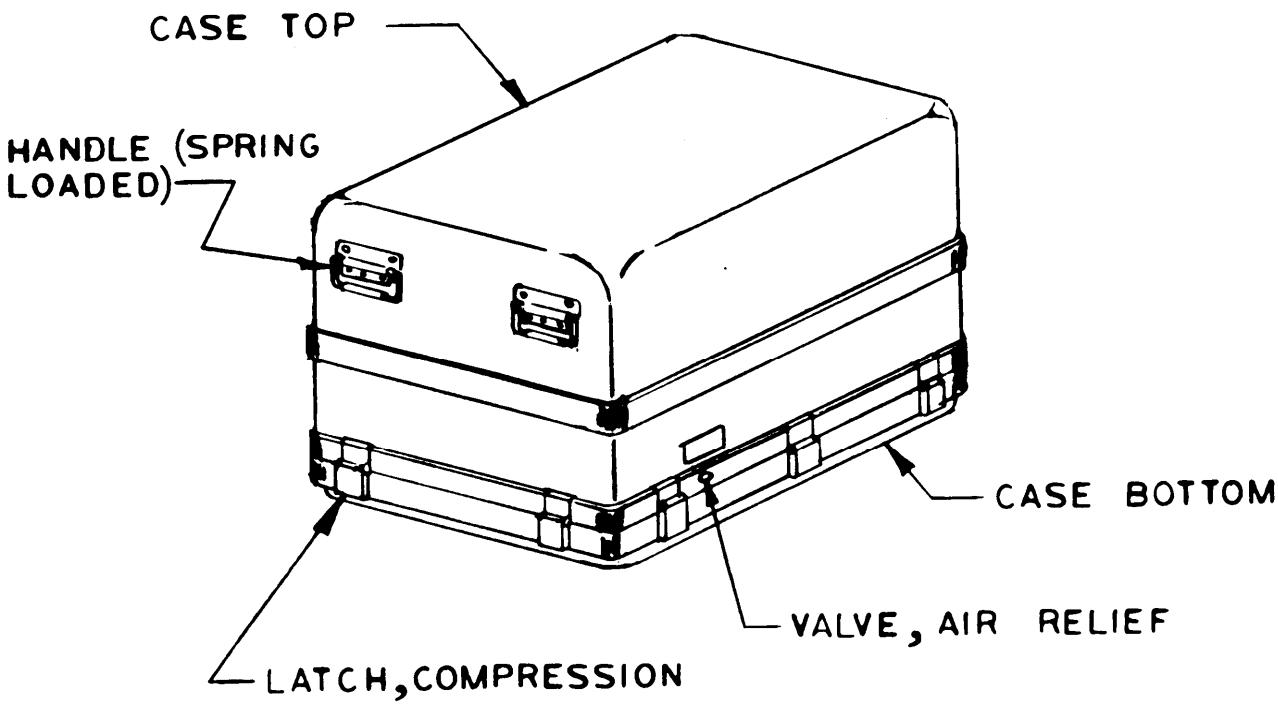


CHAIR AND STOOL

FIGURE 1

SEE 3.4, 3.4.2, 3.4.3 AND 3.17.7.1

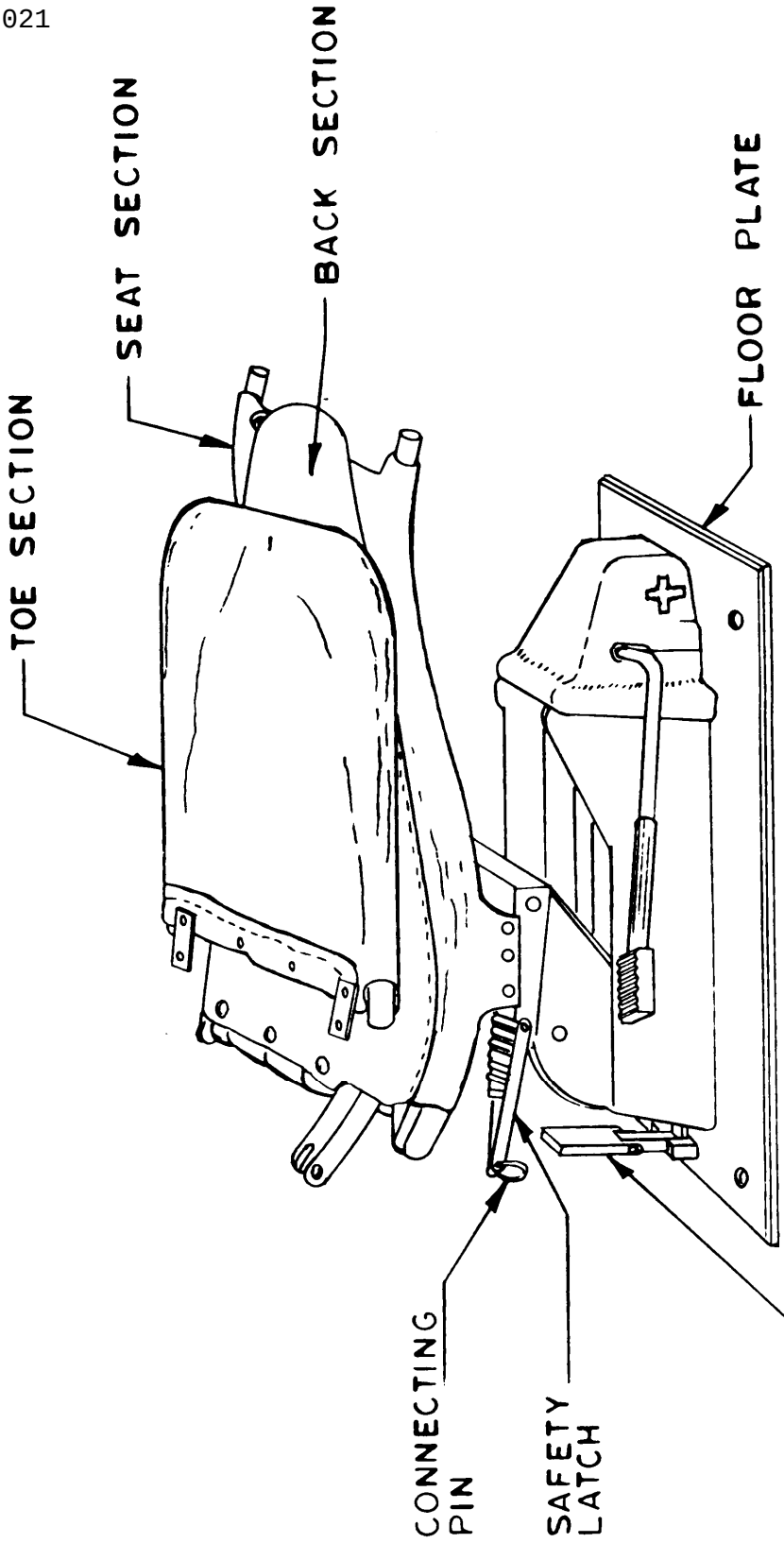
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CARRYING CASE

FIGURE 2 **SEE 3.4**

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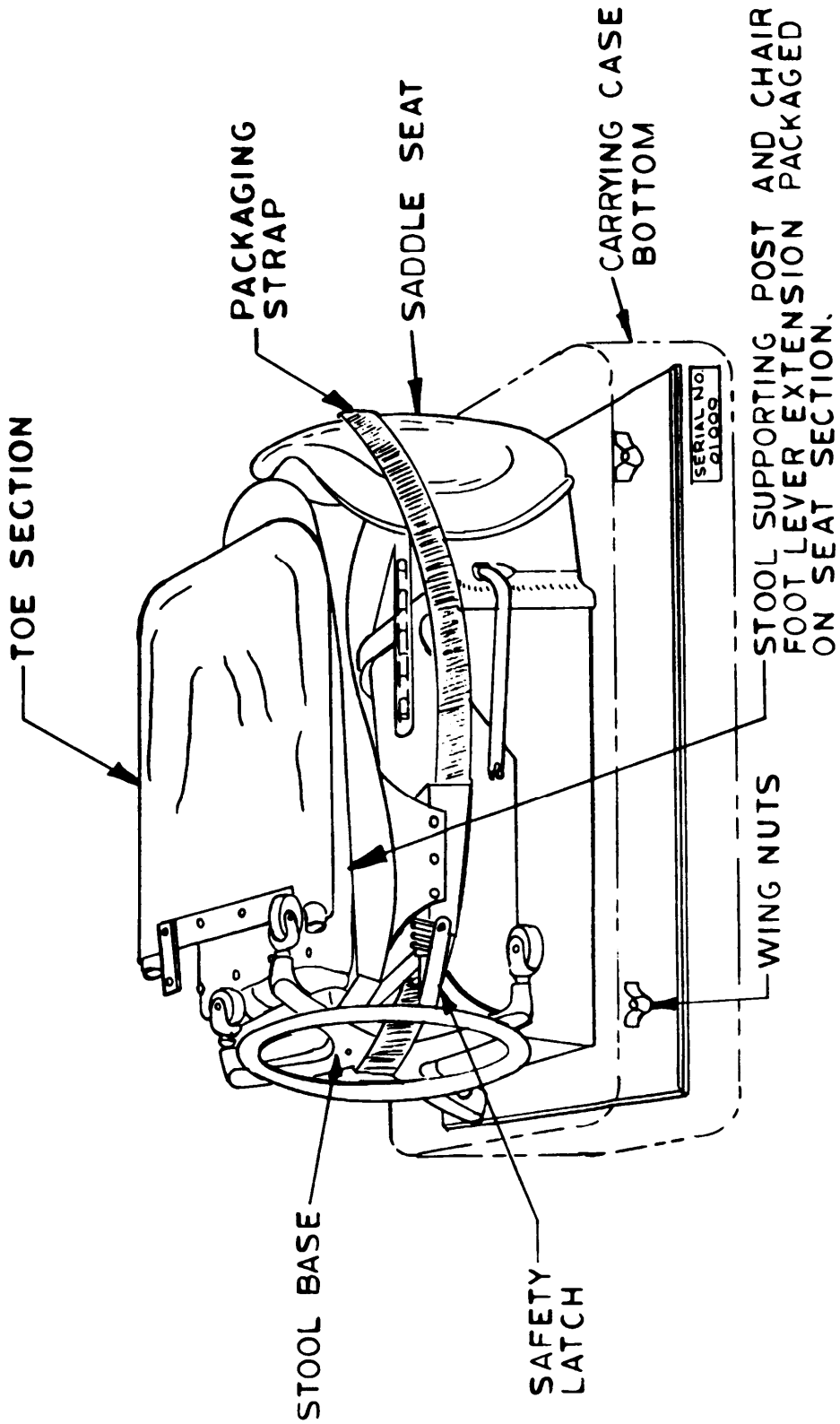


COLLAPSED POSITION

SEE 3.4 AND 3.17.7.2

FIGURE 3

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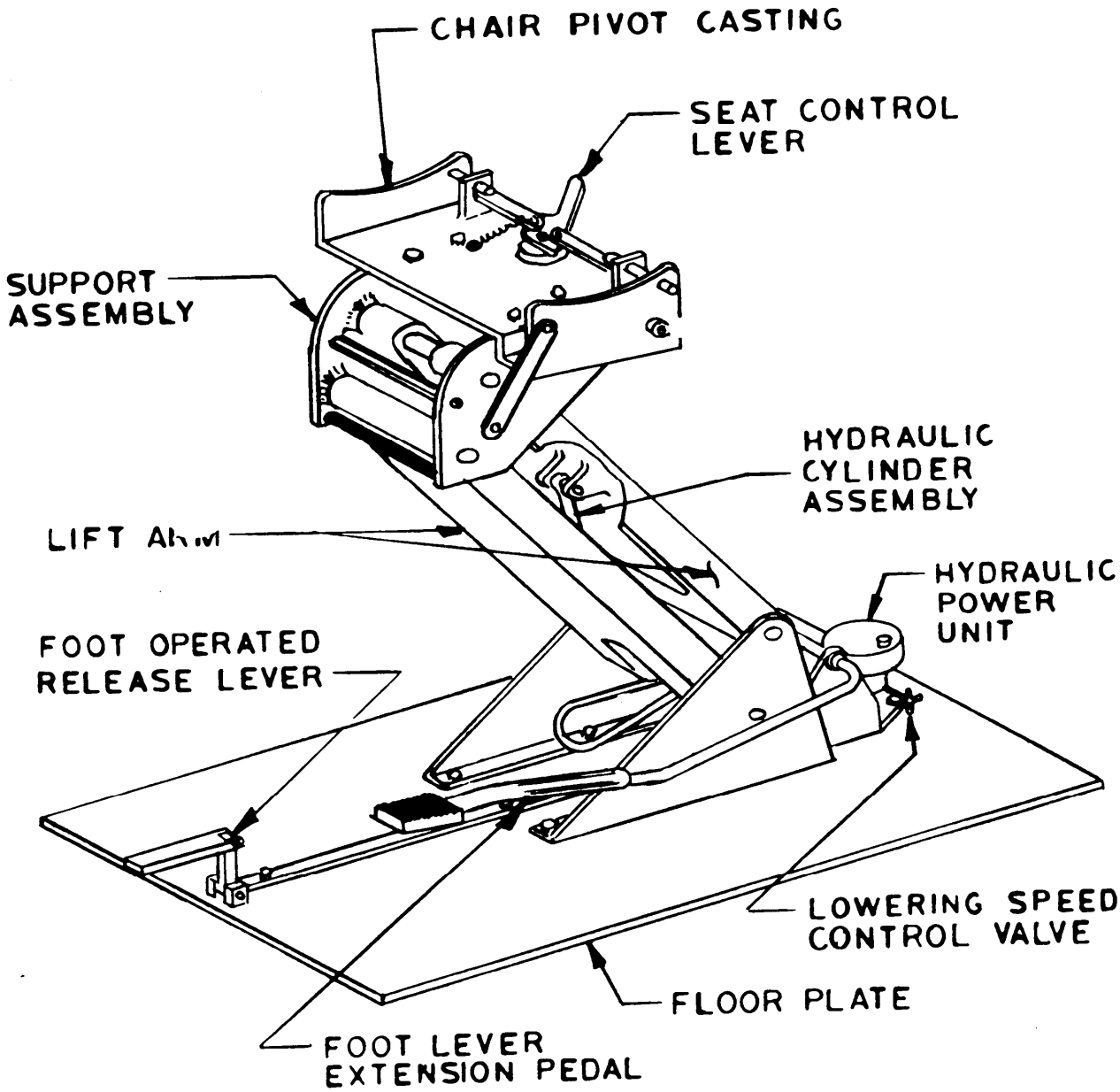


STORAGE POSITION

FIGURE 4

SEE 3.4, 3.4.4 AND 3.17.7.2

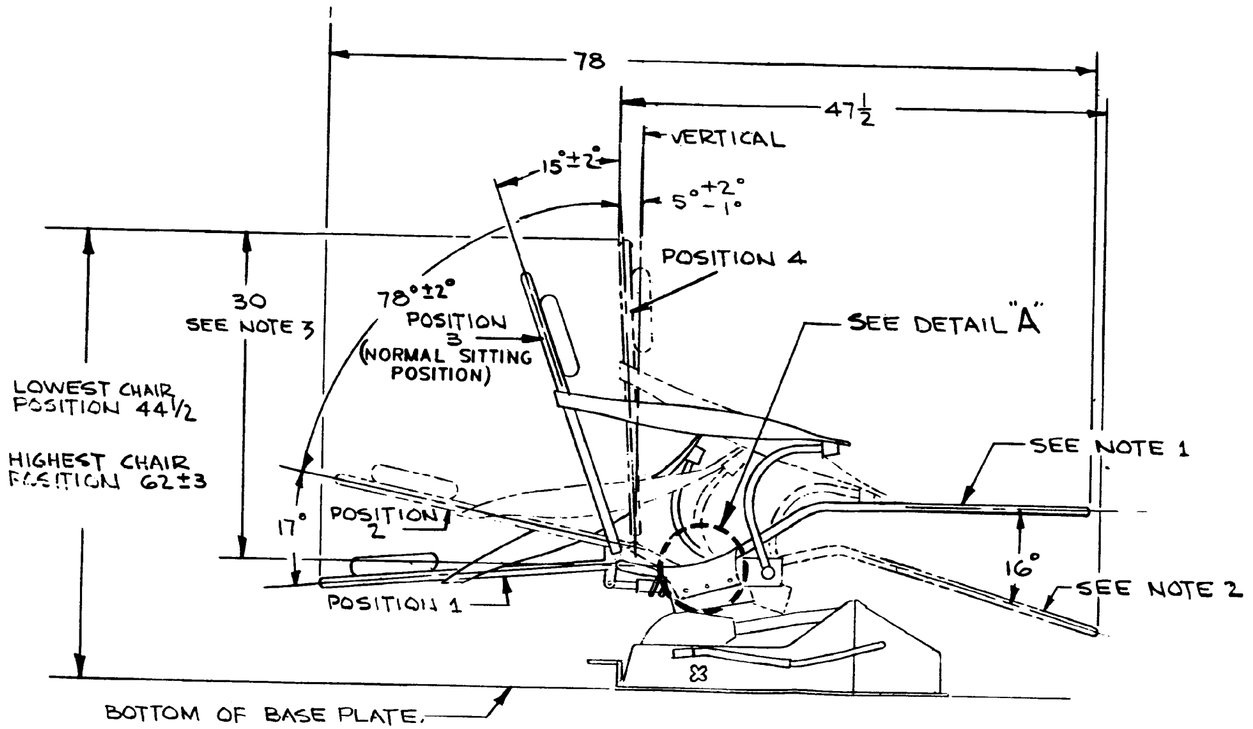
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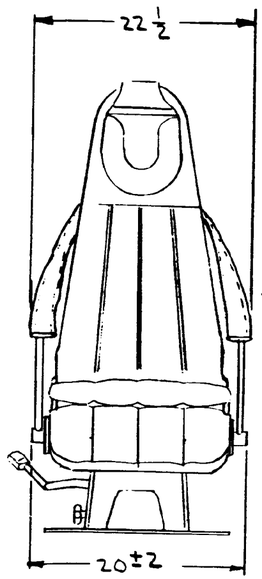


CHAIR BASE
FIGURE 5

SEE 3.4.1

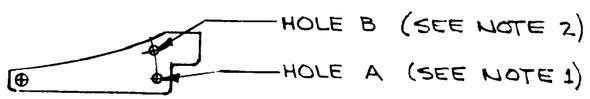
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UNLESS OTHERWISE INDICATED, TOLERANCES SHALL BE:

UNDER 24 IN. ±1 IN.
24 IN. AND ABOVE ±2 IN.
UNDER 20° ±1°
20° AND ABOVE ±2°



DETAIL "A"
SUPPORT, SEAT ADJUSTMENT
REF. DWG. NO. 2534-013

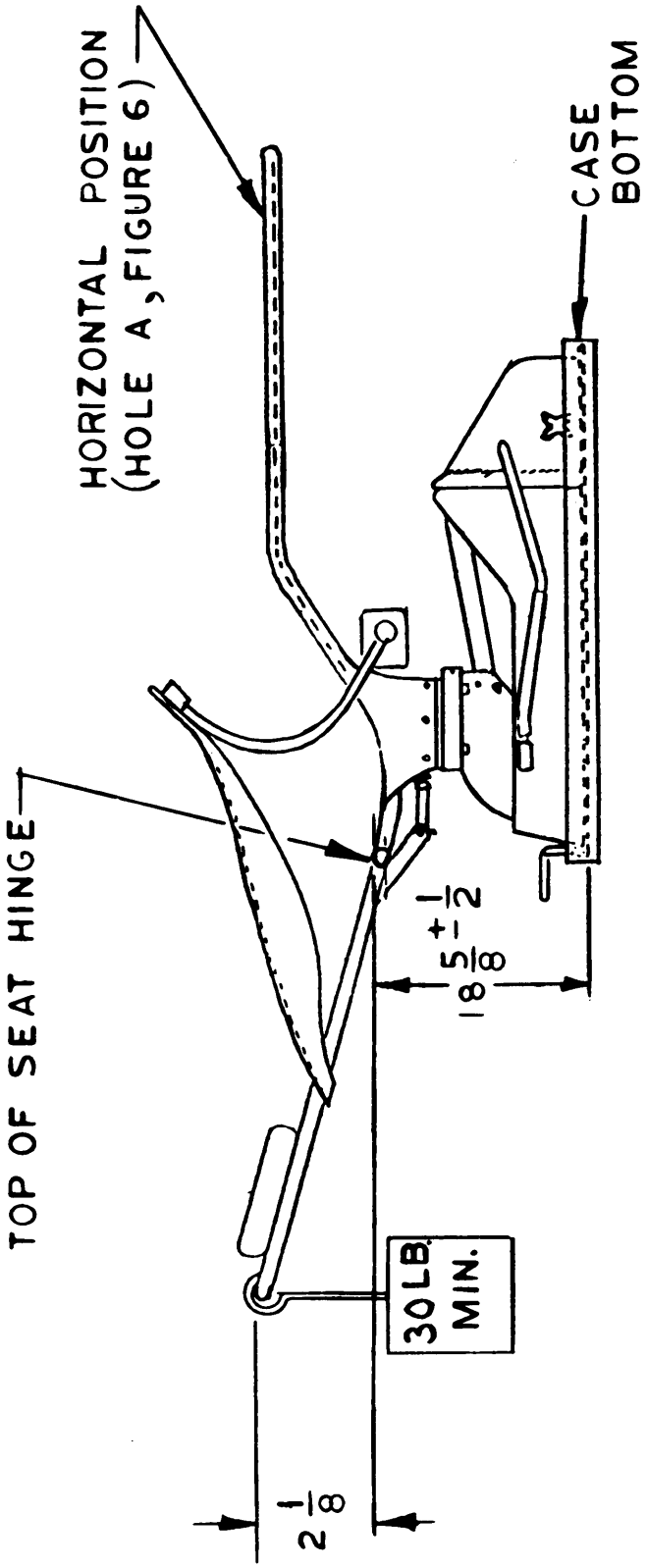
NOTES:

1. SOLID LINES DESIGNATE CHAIR LOCKED IN HOLE A OF "SUPPORT, SEAT ADJUSTMENT."
2. PHANTOM LINES DESIGNATE CHAIR LOCKED IN HOLE B OF "SUPPORT, SEAT ADJUSTMENT."
3. SEAT BACK HEIGHT AS MEASURED FROM TOP OF SEAT AT MAXIMUM CONVEXITY.
4. UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES.

CHAIR-RANGE OF TRAVEL
FIGURE 6

SEE 3.4.2, 3.5.1, 3.17.5.1, 4.5.7.1, 4.5.7.2 AND 6.2.1

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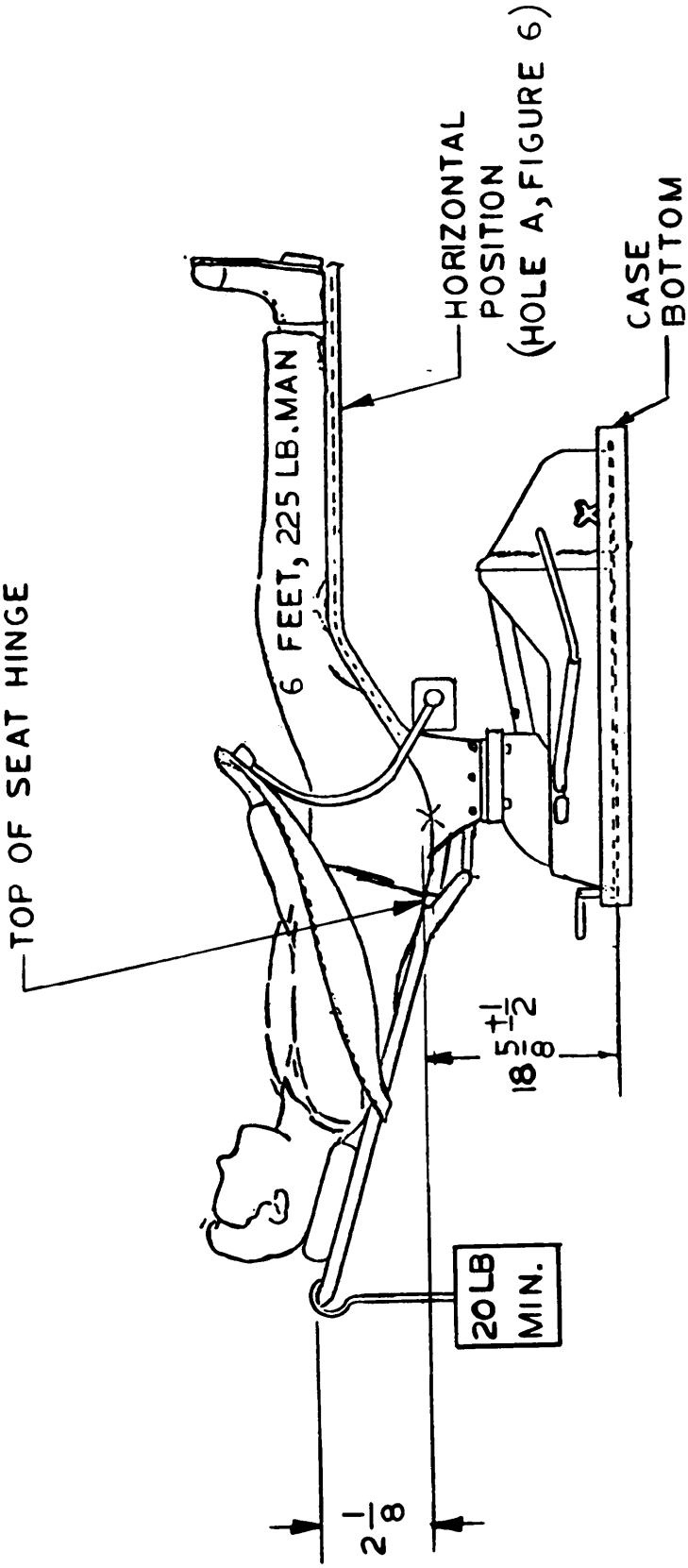


UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES

CHAIR TILT AND DEFORMATION TEST (WEIGHTED
BACK SECTION AND UNLOADED CHAIR) (3.17.1.1)
SEE 4.5.3.1

FIGURE 7

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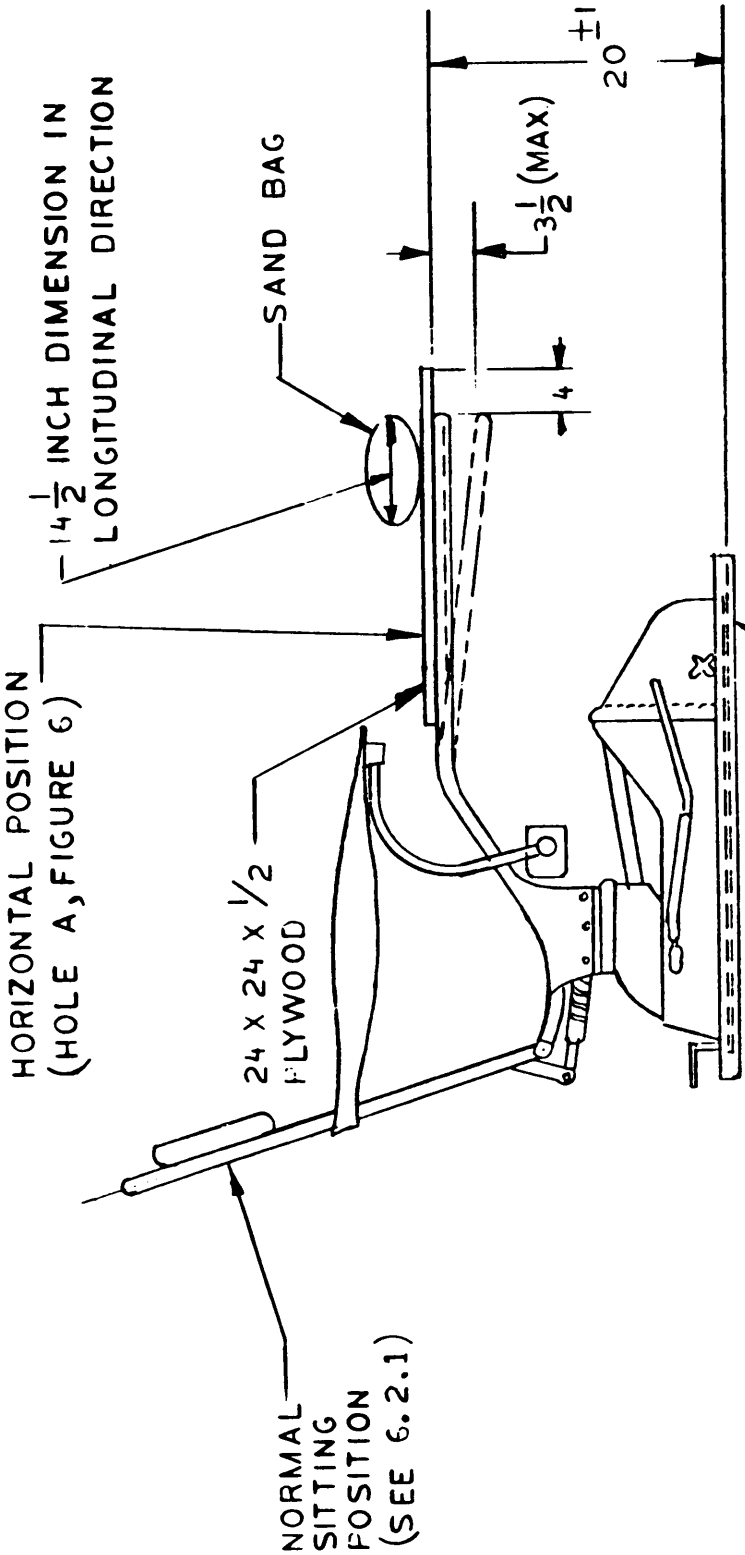


CHAIR TILT AND DEFORMATION TEST (WEIGHTED BACK SECTION AND LOADED CHAIR)

FIGURE 8

(3.17.1.2)
SEE 4.5.3.2

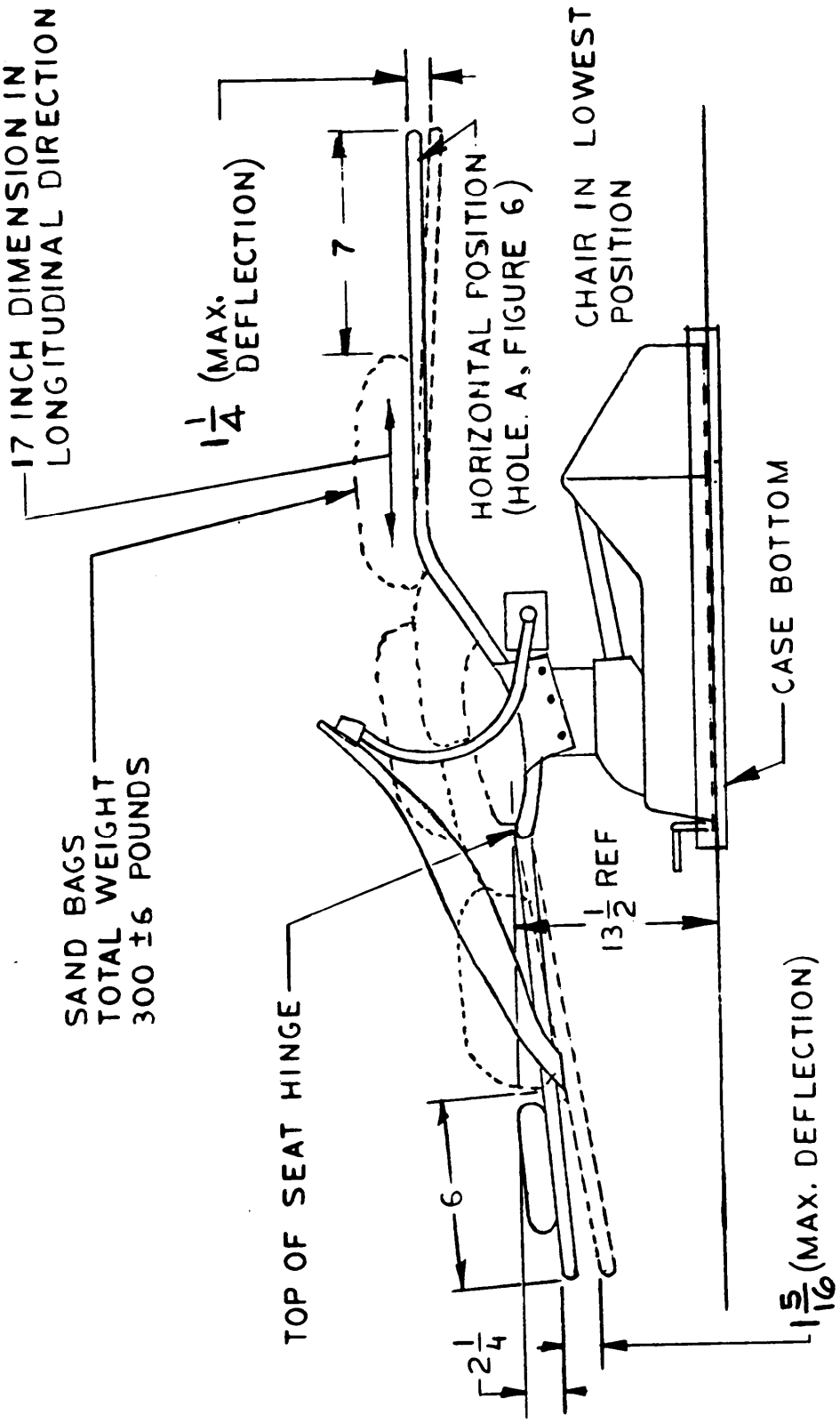
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CHAIR TILT, DEFLECTION AND DEFORMATION
(WEIGHTED TOE SECTION, UNLOADED CHAIR)
(3.17.1.3)
SEE 4.5.3.3 AND 6.22

FIGURE 9

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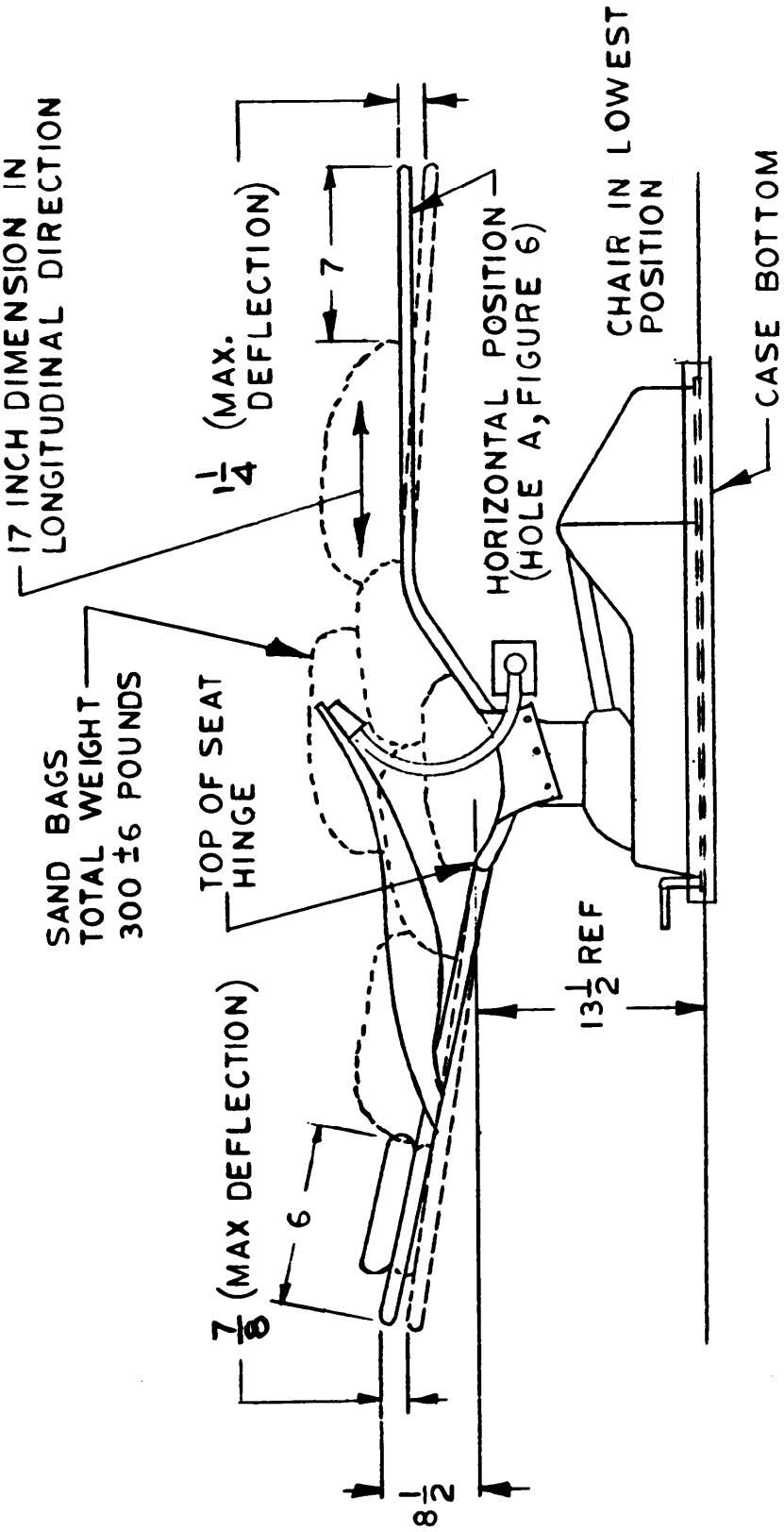


UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.

FIGURE 10 CHAIR STABILITY

SEE 3.17.2, 4.5.4 AND 6.2.2

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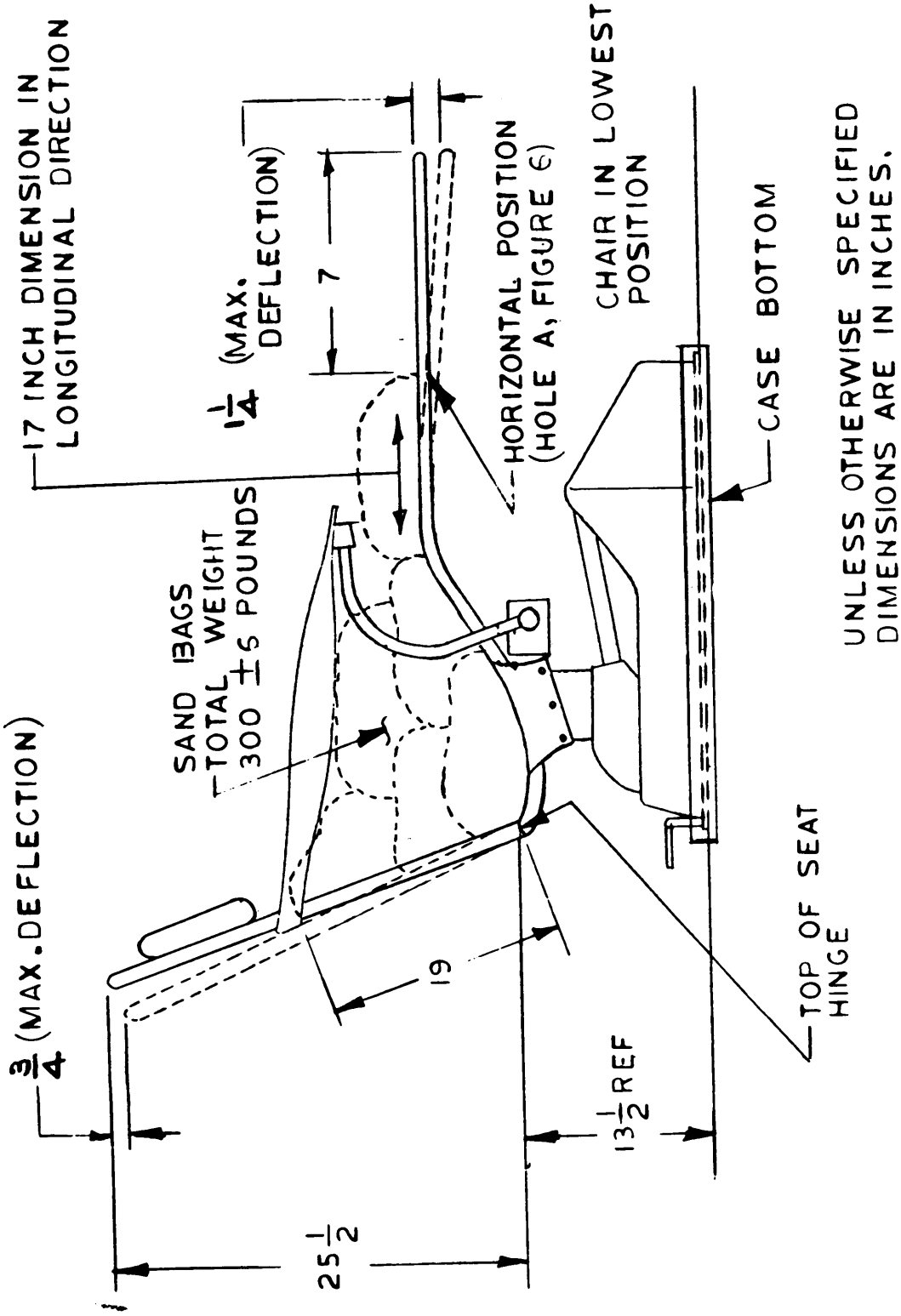
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

CHAIR STABILITY

FIGURE 11

SEE 3.17.2, 4.5.4 AND 6.2.2

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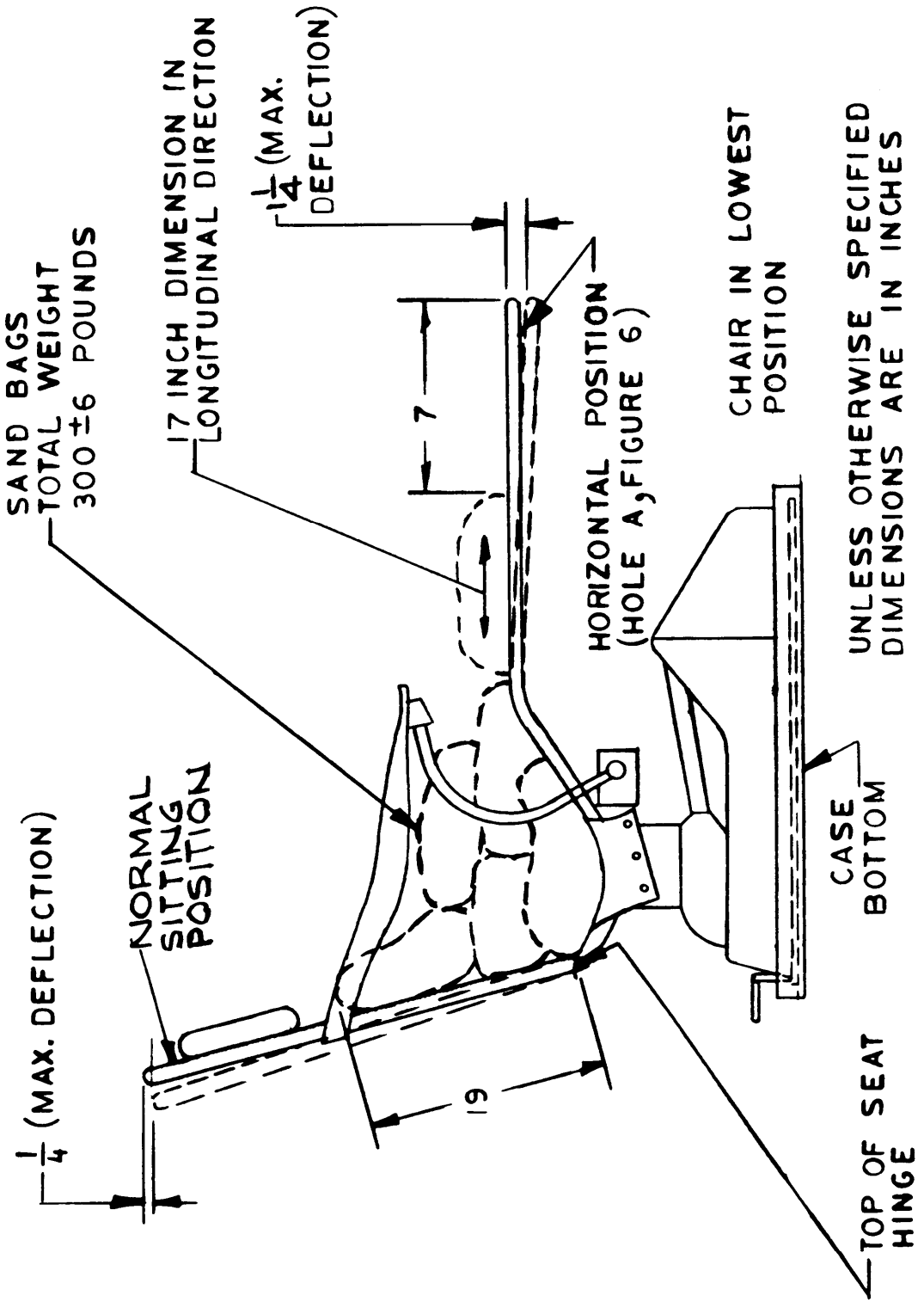


CHAIR STABILITY

FIGURE 12

SEE 3.17.2, 4.5.4 AND 6.2.2

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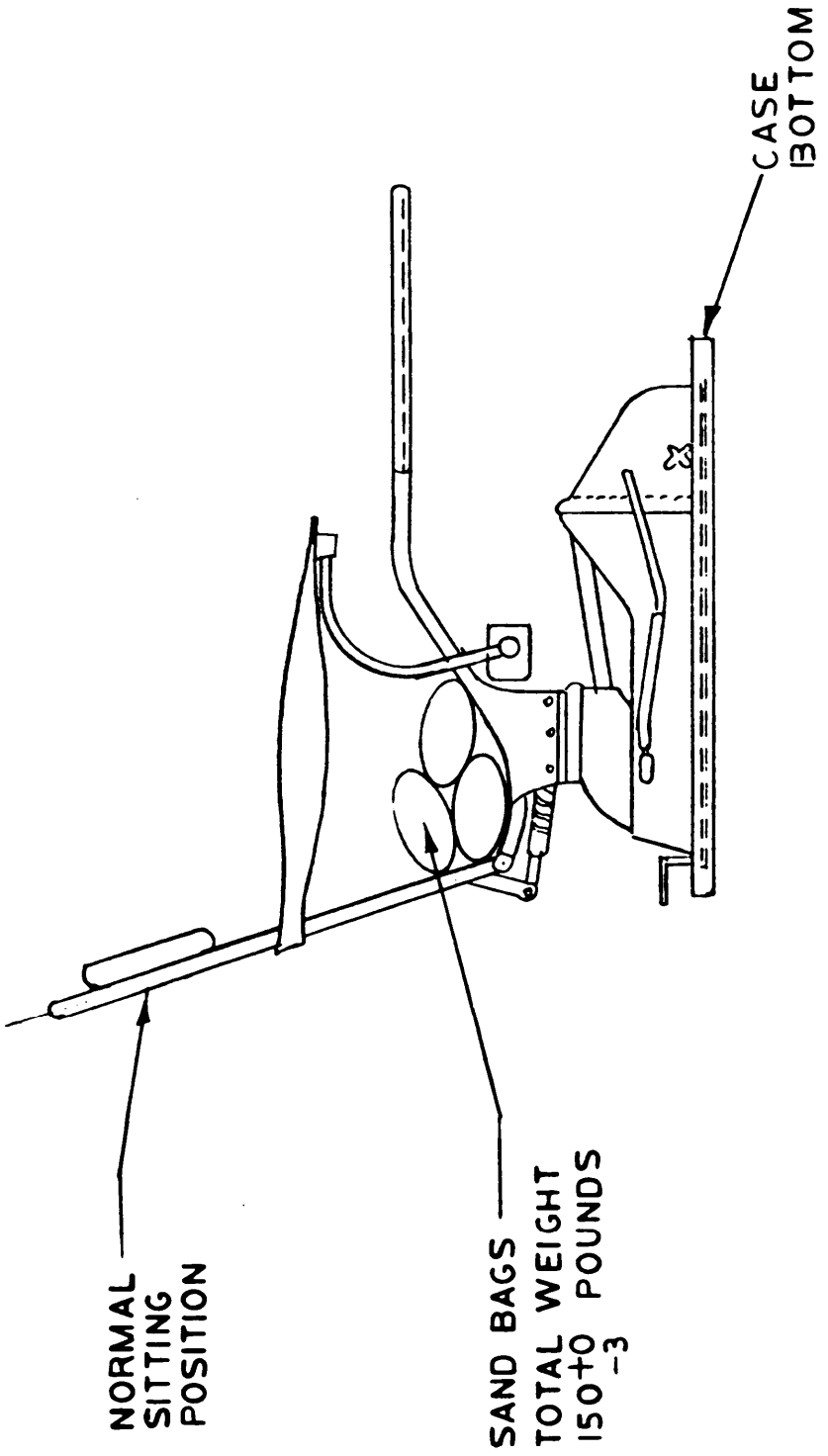


CHAIR STABILITY

FIGURE 13

SEE 3.17.2, 4.5.4, 6.2.1 AND 6.2.2

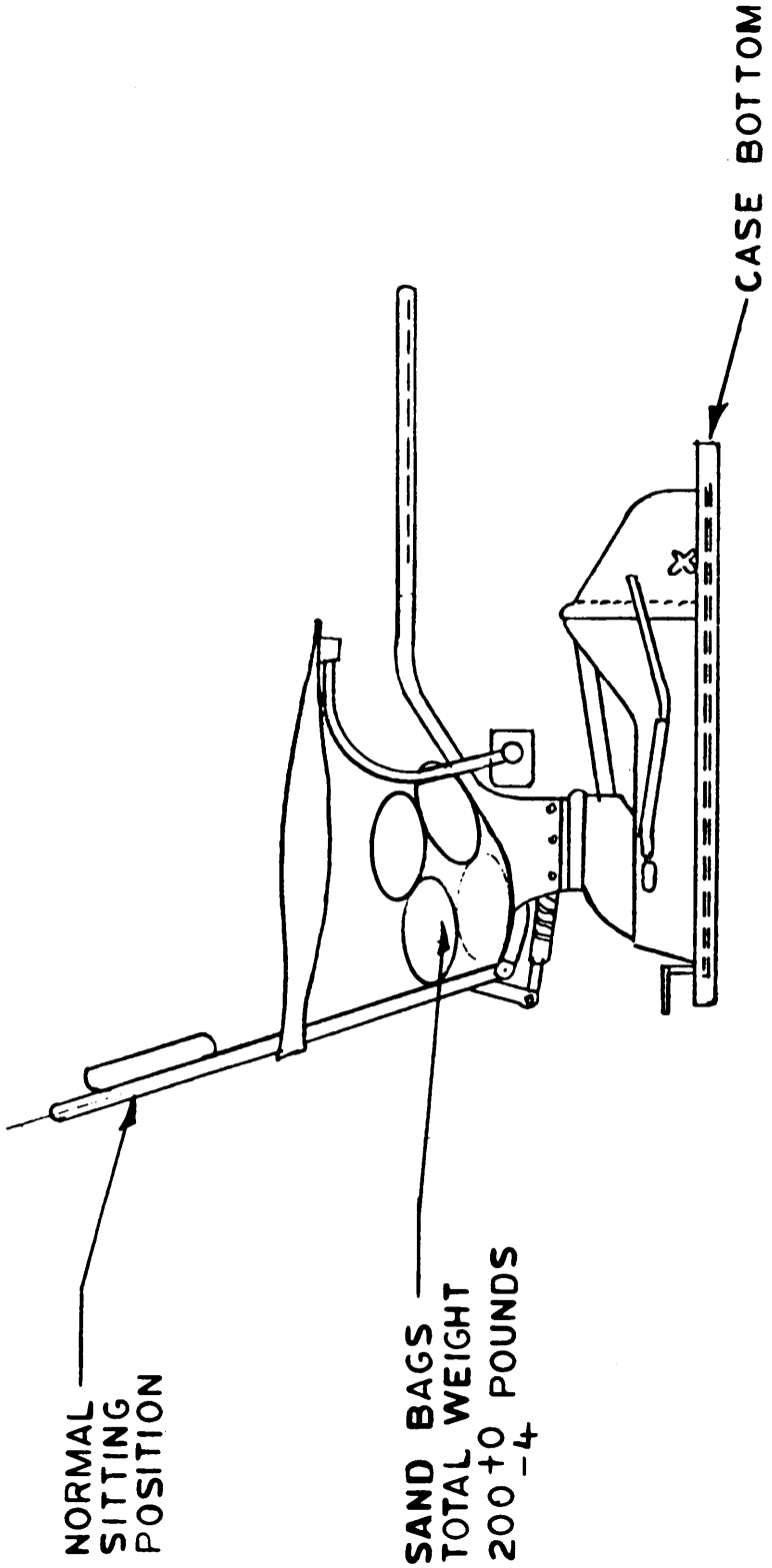
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OPERATION OF THE CHAIR (150 POUND LOAD) (3.17.5.3)
SEE 4.5.7.3, 6.2.1 AND 6.2.2

FIGURE 14

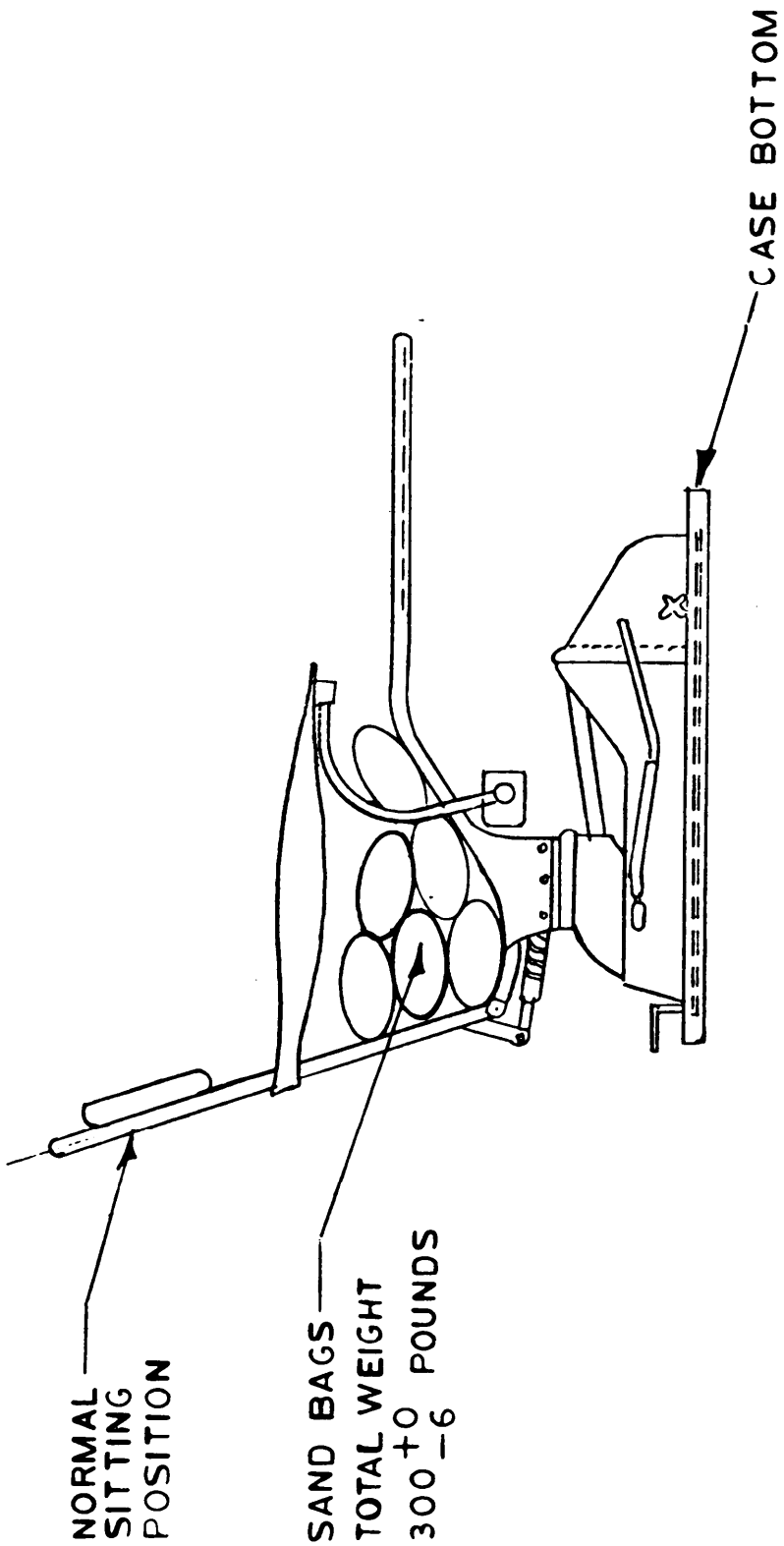
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OPERATION OF THE CHAIR (200 POUND LOAD) (3.17.5.3)
SEE 4.5.7.3, 6.2.1 AND 6.2.2

FIGURE 15

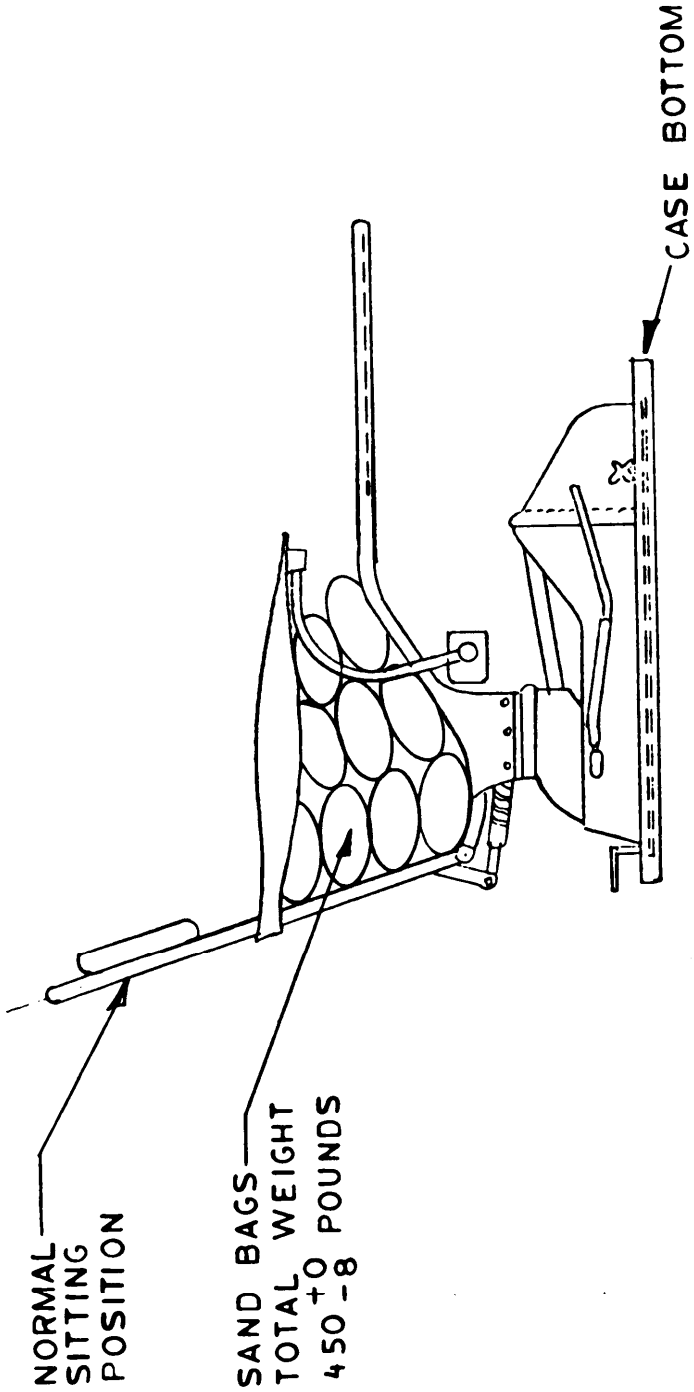
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OPERATION OF THE CHAIR (300 POUND LOAD) (3.17.5.3) SEE 4.5.7.3, 6.2.1 AND 6.2.2

FIGURE 16

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OPERATION OF THE CHAIR (450 POUND LOAD)

FIGURE 17 (3.17.5.3) SEE 4.5.7.3, 6.2.1 AND 6.2.2

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APPENDIX I

TABLE 1 DPSC ASSEMBLY AND SUB-ASSEMBLY PARTS LIST (P/L)

24337	3549-001	
3534-006	3549-002	3638-033
3554-004	3549-003	3638-034
2674-001	3549-004	3638-040
3509-003	3552-004	3638-042
3519-009	3561-009	3639-003
3519-010	3575-002	3644-002
3526-009	3581-004	3669-003
3531-012	3581-005	3673-003
3531-013	3581-006	3673-004
3542-001	3581-007	3676-001
3542-002	3581-008	3685-001
3542-003	3582-002	3685-002
3543-017	3606-007	3686-002
3543-014	3614-007	3686-003
3543-021	3614-008	3686-004
3548-001	3614-009	3686-005
	3614-010	3687-001

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APPENDIX I (cont'd)

TABLE 2 LIST OF DPSC DRAWINGS

"A" SIZE

#1117-00l	#2581-053	#2650-004
#1616-018	#2581-054	#2650-007
#1625-005	#2581-055	#2650-008
#1626-008	#2581-056	#2650-009
#1626-014	#2582-007	#2650-010
#1658-013	#2582-012	#2650-012
#1658-014	#2583-033	#2650-014
#1658-017	#2584-086	#2651-004
#2501-056	#2589-013	#2651-005
#2506-001	#2591-004	#2652-005
#2509-013	#2591-005	#2652-007
#2509-016	#2591-008	#2652-010
#2509-017	#2591-009	#2652-012
#2509-019	#2603-023	#2669-002
#2521-026	#2603-032	#2670-002
#2524-002	#2603-034	#2673-003
#2526-015	#2605-007	#2673-005
#2534-007	#2605-012	#2673-006
#2534-023	#2605-021	#2688-001
#2539-001	#2605-022	#2689-001
#2542-005	#2605-025	#2691-030
#2543-066	#2605-027	#2691-063
#2543-067	#2605-029	#2691-065
#2544-001	#2608-004	#2691-075
#2547-004	#2608-006	*#3543.017
#2547-005	#2608-012	*#3548-001
#2547-006	#2612-008	*#3549-003
#2549-005	#2623-003	*#3549-004
#2549-007	#2623-005	*#3552-004
#2549-008	#2625-002	*#3575-002
#2564-004	#2626-009	*#3581-006
#2564-005	#2633-018	*#3581-008
#2569-001	#2633-023	*#3582-002
#2571-007	#2633-029	*#3608-007
#2573-806	#2633-033	*#3639-003
#2573-007	#2633-034	*#3669-003
#2576-016	#2644-005	*#3676-001
#2577-003	#2644-006	*#3686-002
#2581-052	#2650-003	*#3686-003

* Assembly and Subassembly Drawings

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APPENDIX I (cont'd)

TABLE 2 LIST OF DPSC DRAWINGS (cont'd)

"B" SIZE

#2501-055	#2591-001	*#3543-014
#2501-060	#2593-001	*#3543-021
#2509-014	#2594-008	*#3549-00l
#2509-015	#2598-001	*#3549-002
#2509-025	#2621-001	*#3581-004
#2510-011	#2633-022	*#3581-007
#2519-015	#2633-028	*#3614-008
#2534-004	#2634-004	*#3614-009
#2534-005	#2652-006	*#3614-010
#2534-010	#2691-003	*#3638-031
#2534-011	#2691-005	*#3638-034
#2534-013	#2691-041	*#3638-040
#2543-079	#2691-042	*#3644-002
#2543-082	#2691-060	*#3673-003
#2549-003	#2691-061	*#3673-004
#2549-009	#2693-008	*#3685-001
#2550-012	*#3509-003	*#3685-002
#2563-002	*#3526-009	*#3686-004
#2586-007	*#3531-013	*#3686-005

"C" SIZE

#2515-016	#2581-048	*#3519-009
#2518-001	#2581-049	*#3519-010
#2534-009	#2584-073	*#3531-012
#2564-013	#2633-017	*#3542-00l
#2570-013	#2639-001	*#3542-002
#2570-014	#2644-004	*#3542-003
#2570-015	#2674-001	*#3554-004
#2581-024	#2691-043	*#3561-009
#2581-025	#2691-048	*#3581-005
#2581-031	#2691-062	*#3614-007
#2581-032	#2691-082	*#3638-033
		*#3687-001

"D" SIZE

#2543-076	#2551-038	#2594-007
#2543-080	#2551-043	*#3534-006
#2551-026	#2570-019	*#3638-042
#2551-037	#2594-006	

* Assembly and Subassembly Drawings

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APPENDIX I (cont'd)

TABLE 3 DPSC STANDARD PARTS LIST

1601	Dowel Pin
1603	Spring Pin
1605	Hex Nut
1608	TapTite Screw
1611	Ball
1615	Hexagon Head Cap Screw
1616	Socket Head Cap Screw
1620	Round Head Machine Screw
1622	Flat Head Machine Screw
1623	Pan Head Machine Screw
1624	Truss Head Machine Screw
1625	Set Screw
1626	Shoulder Screw
1627	Pan Head Sheet Metal Screw
1629	Thread Cutting Screw
1630	Set Collar
1637	Pressure Plug
1640	Wing Nut
1645	Socket
1646	stud
1647	Eyelet
1648	Button
1657	Clamps, Electrical
1658	Bushings

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